

Financial Management & Strategic Management

CA Intermediate — Short Revision Notes

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Scope & Objectives of Financial Management

Snapshot

FM plans and controls the firm's financial resources through three (+1) decisions serving one master goal: maximise shareholder **wealth**. Wealth = NPV = present value of expected future *cash* flows discounted for time and risk. Core engine of the whole subject = Time Value of Money.

Core concepts

- **Three decisions:** Investment (capital budgeting + working capital), Financing (capital structure, minimise k), Dividend (payout vs retain). **Fourth silent one:** Liquidity (sub-part of investment; profitability vs liquidity tension).
- **Traditional view (pre-1950s):** finance = procurement of funds only. **Modern view:** procurement AND efficient utilisation (continuous, decision-oriented). ICAI examines modern.
- **3 charges against traditional view:** (i) outsider-looking (only fund suppliers' viewpoint); (ii) over-emphasised episodic long-term financing, ignored working-capital finance; (iii) fund-raising only, ignored allocation/utilisation.
- **Investment decision = most important** — it *creates* value on the asset side; financing/dividend merely redistribute/re-price it.

Key provisions / rules

Decision → tool: Investment(LT)→NPV/IRR/PI/payback; Investment(ST)→operating cycle, liquidity-profitability; Financing→WACC, leverage, EBIT-EPS; Dividend→Walter/Gordon/MM, payout ratio.

Profit vs Wealth maximisation — profit fails on 4 counts: **Ambiguous** (PBT/PAT? total/per share? short/long run) · **Ignores timing** (time value) · **Ignores risk** · **Ignores cash quality** (profit = opinion, cash = fact). Wealth fixes all four. Profit = *necessary* condition for survival but *wrong objective*. EPS maximisation still fails on timing/risk/dividend and is manipulable. ICAI objective = **shareholder** wealth (not total firm value) within ethical/stakeholder constraints (constraints, not competing objectives).

Discount rate decomposition: $k \approx \text{real risk-free rate} + \text{inflation premium} + \text{risk premium}$.

Concept	Formula
NPV (wealth)	$NPV = \sum [C_t \div (1+k)^t] - C_0$; accept if $NPV > 0$
FV single sum	$FV = PV \times (1+i)^n$ (factor = FVIF)
PV single sum	$PV = FV \div (1+i)^n$ (factor = PVIF)
Intra-year compounding	$FV = PV \times (1 + i/m)^{(m \times n)}$
Effective annual rate	$EAR = (1 + i/m)^m - 1$; continuous → $e^i - 1$
FV ordinary annuity	$FVA = A \times [((1+i)^n - 1) \div i]$ (FVIFA)
PV ordinary annuity	$PVA = A \times [(1 - (1+i)^{-n}) \div i]$ (PVIFA)
Annuity due	multiply ordinary result $\times (1+i)$
Deferred annuity	$PV = A \times PVIFA(i,n) \times PVIF(i,d)$; d = deferment
Perpetuity	$PV = A \div i$
Growing perpetuity	$PV = A_1 \div (i - g)$; needs $i > g$
Sinking fund (deposit)	$A = FV \times [i \div ((1+i)^n - 1)] = FV \div FVIFA$
Loan instalment / capital recovery	$A = PV \times [i \div (1 - (1+i)^{-n})] = PV \div PVIFA$
Rule of 72	doubling years $\approx 72 \div \text{rate\%}$ (quick check only)

- **Sinking-fund factor = $1/FVIFA$; capital-recovery factor = $1/PVIFA$** — memorise 2 factors, invert when instalment is unknown.
- **Amortisation:** interest = opening balance $\times i$ (on *declining* balance); principal = instalment – interest; total interest = $(n \times \text{instalment}) - \text{principal}$.
- **Agency:** owner(principal) vs manager(agent) $\rightarrow$ agency costs = **monitoring** (principal) + **bonding** (agent) + **residual loss**. NOT the manager's salary. Fixes: ESOPs/incentive pay, board & audit, debt covenants, takeover threat, SEBI. **Second conflict:** shareholders vs debt-holders (asset substitution, excess dividends) $\rightarrow$ curbed by covenants.
- **Treasurer** (outward/forward: funds, cash, credit, investment, insurance) vs **Controller** (inward/backward: accounting, audit, budgeting, tax, internal control). Both under CFO.

Worked mini-example

Retain-or-pay (₹10,00,000 surplus; shareholders' opportunity rate $k = 12\%$). Retain only if reinvestment $r > k$.

- Scenario A: $r = 15\% \rightarrow ₹11,50,000$ in 1 yr $\rightarrow PV = 11,50,000 \times 0.8929 = ₹10,26,835 > ₹10,00,000 \rightarrow$ **Retain** (+ ₹26,835).
- Scenario B: $r = 9\% \rightarrow ₹10,90,000 \rightarrow PV = ₹9,73,261 < ₹10,00,000 \rightarrow$ **Pay out** (–₹26,739).
- Break-even $r = k = 12\%$: retention $\rightarrow ₹11,20,000 \rightarrow PV = ₹10,00,000$ (indifference).

Exam traps & must-remember

1. "Beginning of each year" = **annuity due** $\rightarrow \times (1+i)$. Most common lost mark.
2. "12% compounded quarterly" $\neq 12\%$ effective $\rightarrow$ use i/m and $m \times n$, or EAR. Monthly EMIs: $m=12$.
3. **FVIFA vs PVIFA:** accumulate to future target $\rightarrow FVIFA$ (sinking fund); value stream today / loan instalment $\rightarrow PVIFA$.
4. NPV uses **cash flows**, not accounting profit (add back depreciation).
5. Different risk $\Rightarrow$ different discount rate. Never lazily apply one rate to two projects.
6. "Equal total cash $\Rightarrow$ equal projects" — false; timing & risk differ; always discount.
7. Perpetuity A/i only for "forever"; fixed years $\rightarrow PVIFA$.
8. Growing perpetuity needs $i > g$; if $g \geq i$ formula invalid.
9. Wealth maximisation $\neq$ share-price manipulation; it's *sustainable* value within ethics/stakeholders.
10. Agency cost = cost of conflict + control (monitoring/bonding/residual), NOT salary.
11. **Deferred annuity:** PVIFA reaches one period *before* first flow, then extra PVIF discount.
12. Amortisation interest on **declining** balance (flat-rate error overstates interest).
13. Profit maximisation: state it's *necessary but insufficient*, not "stupid."
14. EAR ceiling = $e^i - 1$ (continuous); more frequency = diminishing returns.
15. Objective = **shareholder** wealth, not total firm value (matters when value shifts lender $\leftrightarrow$ owner).

One-line recall

- Wealth = $\sum C_t / (1+k)^t - C_0$; beats profit on timing, risk, ambiguity, cash quality.
- Three decisions: invest $\cdot$ finance $\cdot$ dividend (+ liquidity); investment decision creates value.
- Compound to go forward, discount to come back; sinking-fund & loan factors are inverted annuity factors.
- Retain only if $r > k$; else distribute.
- Agency cost = monitoring + bonding + residual loss; align via ESOPs, covenants, takeover threat.
- Amortisation: interest = opening $\times i$ on declining balance; principal = instalment – interest.

Types of Financing & Financial Markets

Snapshot

The financing decision picks the mix of sources leaving shareholders richest and firm safest. Every source carries four attributes: **Cost, Risk, Control, Maturity**. Financing merely *slices* the pie — it changes firm value only through frictions: tax shield, distress/bankruptcy cost, flotation cost, information asymmetry, agency cost.

Core concepts

- **Queue of claims:** Debt (first, lowest cost, tax-deductible) → Preference (middle) → Equity (residual, last, highest cost). Higher risk demands higher return.
- **Equity is the MOST expensive source**, not the cheapest (16-18% vs 9-11% debt); return arrives as expected dividends + capital gains, not a cheque.
- **3 reasons debt is cheaper:** (i) priority of claim (lower risk); (ii) security/charge on assets; (iii) **tax shield** (pure govt subsidy, not risk) → $K_d = i \times (1 - t)$. Shield worthless if no taxable profit.
- **Classification lenses:** by maturity (long/medium/short), ownership (owners' vs borrowed), origin (internal vs external), effort (negotiated vs spontaneous).

Key provisions / rules

Long-term owners' funds: Equity (permanent, votes/control, dilutes, highest cost, no tax deduction, no charge) · Preference (fixed dividend before equity, priority in repayment, usually no vote — regain if unpaid 2 yrs; **redeemable within 20 yrs** under Companies Act 2013, infra excepted; cumulative default) · Retained earnings (internal; opportunity cost = shareholders' required return, NOT free; marginally cheaper than fresh equity only by saving flotation cost).

- Capital order: Authorised ≥ Issued ≥ Subscribed ≥ Called-up ≥ Paid-up. Dividends/bonus on **face** value. Securities premium = capital reserve, not distributable.
- **Bonus issue** raises NO new cash (capitalises reserves); **rights issue** does.

Long-term borrowed: Debentures (fixed interest paid regardless of profit, tax-deductible = cheapest; no vote; often secured; FCD/PCD/NCD, zero-coupon at discount) · Term loans (covenants, secured) · Venture capital (new high-risk ventures; stages: Seed → Start-up → Early → Expansion → Later/Mezzanine/Bridge; demands 30-50%+ for portfolio failures) · ADR (US equity) / GDR (non-US equity) / ECB (foreign-currency **debt**, currency risk) / FDI (control) vs FII/FPI (portfolio) / Securitisation (off-balance-sheet).

- Fixed charge (specific asset) ranks ahead of floating charge (changing pool, crystallises on default) on charged assets.

Medium-term: Lease (lessor owns & claims depreciation; lessee deducts rental only) — **Finance/Capital lease** (non-cancellable, transfers risks/rewards, on lessee's books) vs **Operating lease** (cancellable, lessor bears obsolescence). Sale-and-lease-back. Hire purchase (ownership transfers after last instalment; hirer claims depreciation, deducts interest part only). Public deposits (6mth-3yr).

Short-term: Trade credit (spontaneous, free unless discount forgone) · Cash credit (against drawing power on hypothecated stock) · Overdraft (fluctuating gaps) · Bills discounting · Letter of credit · Commercial Paper (unsecured PN, 7d-1yr, at discount, high-rated only) · Factoring (sell receivables; with/without recourse; Forfaiting = export, always without recourse) · Accruals (wages/tax payable — cheapest, zero cost, non-manoeuvrable).

Purpose	Formula
After-tax cost of debt	$K_d = \text{Interest rate} \times (1 - t)$
EPS (financing mix)	$\text{EPS} = \frac{[(\text{EBIT} - \text{Interest}) \times (1 - t) - \text{Pref. Div}]}{\text{No. of equity shares}}$
Favourable leverage	Return on capital $(\text{EBIT} \div \text{Total capital}) > \text{fixed-charge rate}$
Indifference EBIT	set $\text{EPS}(\text{Plan1}) = \text{EPS}(\text{Plan2})$, solve

Purpose	Formula
Lease vs Buy	choose lower PV of after-tax cash outflows
Buy outflow/yr	after-tax interest + principal – depreciation tax shield (– salvage)
Depreciation tax shield	Depreciation × t
After-tax lease rental	Lease rent × (1 – t)
Cost of forgoing discount	$[d \div (100 - d)] \times [365 \div (\text{Credit period} - \text{Discount period})]$
Ex-rights price	$(N \times \text{Cum-rights price} + \text{Issue price}) \div (N + 1)$
Value of a right	$(\text{Cum-rights price} - \text{Issue price}) \div (N + 1)$

Markets: Money market (≤ 1 yr, RBI; T-Bills [Govt, zero-coupon], CP [companies], CD [banks], Call money [≤ 1 day]/Notice [2-14 days], Repos, Commercial Bills). Capital market (> 1 yr, SEBI; shares/debentures/bonds). Primary = new issues, **company gets cash** (IPO/FPO/Rights/Private placement/Preferential). Secondary = existing securities, no cash to firm but provides **liquidity + price discovery** (which cheapens future primary fund-raising).

Matching (hedging) principle: match maturity of source to life of asset. Permanent (core) current assets → long-term funds; temporary (seasonal) → short-term. **Aggressive** (short funds also fund permanent needs: high return/high liquidity risk) · **Hedging** (balanced) · **Conservative** (long funds fund temporary: low return, idle cost/over-capitalisation).

Worked mini-example

Trade credit "2/15, net 60", purchases ₹40,00,000/yr, bank rate 14%. Cost = $[2 \div 98] \times [365 \div (60-15)] = 0.020408 \times 8.111 = 16.55\% \text{ p.a.} > 14\% \rightarrow \text{take the discount, borrow if needed}$. Rupee gain $\approx$ discount $2\% \times 40,00,000 = ₹80,000$ – interest ($\approx 39,20,000 \times 14\% \times 45/365 \approx ₹67,660$) $\approx ₹12,340/\text{yr}$.

Exam traps & must-remember

- "Equity is free/cheapest" — wrong, it's **most expensive** (no contractual cheque $\neq$ free).
- Retained earnings have **opportunity cost** = K_e ; only marginally cheaper via saved flotation.
- Cost of debt = **after-tax**; preference dividend gets **no** shield; shield worthless without taxable profit.
- Preference dividend deducted **after** tax (appropriation of PAT); interest before tax.
- Company gets cash only in **primary** market; secondary still matters (liquidity/price discovery).
- Money vs capital market cut-off = **1 year**. CP & CD are money-market (not capital); T-Bills = Govt/RBI.
- Hire purchase → ownership transfers; lease → lessor keeps ownership. Depreciation claimed by owner.
- Matching: long assets $\leftrightarrow$ long funds. Short-fund-long-asset = liquidity risk; long-fund-short-asset = idle cost.
- "More debt always raises EPS" — only above indifference EBIT (favourable leverage); below it EPS falls/negative.
- Bonus issue = no new cash; rights issue = cash.
- Redeemable preference within **20 years** (infra excepted); no irredeemable preference in India.
- Trade credit "free" only until discount forgone ($\sim 16.5\%$ for 2/15 net 60).
- ADR/GDR = equity abroad; **ECB = debt + currency risk**.
- Skipped cumulative preference dividend = **arrears**; skipped debenture interest = **default/insolvency**.
- Without-recourse factoring: bad-debt risk moves to factor (bad-debt saving only here).
- Ignoring a right forfeits its value; rights issue wealth-neutral only if you act.

One-line recall

- Four attributes: cost, risk, control, maturity; financing only slices the pie (frictions give it teeth).
- Queue: debt < preference < equity in cost; debt cheapest via priority + security + tax shield.
- $K_d = i(1-t)$; $\text{EPS} = [(EBIT - I)(1-t) - \text{Pref div}] \div \text{shares}$.
- Match source maturity to asset life; aggressive/hedging/conservative spectrum.
- Money market $\leq 1\text{yr}$ (RBI); capital market $> 1\text{yr}$ (SEBI); cash only in primary market.
- Factor/take-discount/lease decisions: compare cost vs benefit in PV terms.

Ratio Analysis (Financial Analysis & Planning)

Snapshot

A ratio = one figure ÷ another chosen so the answer means something (manufactures a yardstick). Ratios are scale-free and compared against three benchmarks: own past (trend), rival (cross-section), industry norm. Organised by four families serving the three FM decisions. No single "correct" value — context-dependent.

Core concepts

- **Four forms:** proportion "x:1" (same kind of thing), "%" (return/part-of-whole), "times" (coverage/velocity), "days" (duration). Times × days = period; turnover and holding period are reciprocals scaled by period.
- **Stock ratio** (two balance-sheet items, no averaging) vs **flow-to-stock ratio** (P&L flow over balance stock → use **average of opening & closing** denominator).
- **Liquidity-profitability trade-off** is why a *panel* of ratios is needed; high liquidity lowers return.
- Reading direction: Debt-Equity, Capital Gearing, Debt Ratio → higher = riskier. Proprietary + all coverage ratios → higher = safer.

Key provisions / rules

#	Ratio	Formula	Ideal/sense
Liquidity			
1	Current	Current Assets ÷ Current Liabilities	2:1
2	Quick/Acid	(CA – Inventory – Prepaid) ÷ CL	1:1
3	Cash/Absolute	(Cash+Bank+Marketable sec) ÷ CL	0.5:1
—	Net Working Capital	CA – CL	positive cushion
Solvency/Leverage			
4	Debt-Equity	Long-term Debt ÷ Shareholders' Funds	≤2:1
5	Proprietary	Shareholders' Funds ÷ Total Assets	higher safer
6	Capital Gearing	(Pref + Debt) ÷ (Equity cap + Reserves)	context
7	Interest Coverage	EBIT ÷ Interest	higher safer
—	Debt to Total Assets	Total Debt ÷ Total Assets = 1 – Proprietary	—
—	DSCR	(PAT + Dep + Int) ÷ (Int + Principal)	~1.5-2
—	Fixed Charges Coverage	(EBIT + fixed charges pre-tax) ÷ (Int + fixed charges + Pref div grossed up)	—
Activity/Turnover			
8	Inventory Turnover	COGS ÷ Avg Inventory	higher=leaner
9	Inventory period	360 ÷ Inventory Turnover	fewer days
10	Debtors Turnover	Credit Sales ÷ Avg Receivables	higher
11	Collection period	360 ÷ Debtors Turnover	fewer days
12	Creditors Turnover	Credit Purchases ÷ Avg Payables	context
13	Payment period	360 ÷ Creditors Turnover	longer=free finance
14	Operating Cycle	Inventory period + Collection period	fewer
15	Cash Conversion Cycle	Inv + Debtor – Creditor period	fewer
16-19	Fixed Asset / Total Asset / Capital / WC Turnover	Sales ÷ (Net FA / Total Assets / Capital Employed / NWC)	higher
Margins (÷Sales)			

#	Ratio	Formula	Ideal/sense
20	Gross Profit	Gross Profit ÷ Sales	higher
21	Operating Profit	EBIT ÷ Sales	higher
22	Operating Ratio	(COGS + Op exp) ÷ Sales	Op Ratio + Op Profit Ratio = 100%
23	Net Profit	PAT ÷ Sales	higher
Returns (÷Capital)			
24	ROCE	EBIT ÷ Capital Employed	higher
25	ROE / Return on Net Worth	PAT ÷ Shareholders' Funds	higher
26	ROA	PAT (or EBIT) ÷ Total Assets	higher
—	Sustainable Growth Rate	ROE × Retention (b)	growth w/o new equity
Valuation (÷share)			
27	EPS	(PAT – Pref Div) ÷ No. equity shares	higher
28	P/E	Market Price ÷ EPS	growth signal
29	Dividend Payout	DPS ÷ EPS	policy
30	Retention	1 – Payout	policy
31	Dividend Yield	DPS ÷ MPS	income
32	Earnings Yield	EPS ÷ MPS = 1 ÷ (P/E)	valuation
33	Book Value/Share	(Equity cap + Reserves) ÷ equity shares	—
—	Dividend Cover	EPS ÷ DPS = 1 ÷ Payout	higher safer
—	Market-to-Book	MPS ÷ Book Value/Share	>1 = value created

- **Capital Employed** = Total Assets – Current Liabilities = Shareholders' Funds + Long-term Debt.
- **Quick Assets** = CA – Inventory – Prepaid. **Shareholders' funds** = Equity cap + Reserves + Preference cap – fictitious assets.
- **DuPont 3-step:** $ROE = (PAT/Sales) \times (Sales/Total\ Assets) \times (Total\ Assets/Shareholders'\ Funds) = Net\ Margin \times Asset\ Turnover \times Equity\ Multiplier$ (levers = operating/pricing, investing, financing).
- **DuPont 5-step:** $ROE = (PAT/PBT) \times (PBT/EBIT) \times (EBIT/Sales) \times (Sales/Assets) \times (Assets/Equity) = Tax\ burden \times Interest\ burden \times Op\ margin \times Asset\ TO \times Leverage$.
- **ROCE 2-factor:** $ROCE = Capital\ Turnover \times Operating\ Margin$.
- Manufacturing operating cycle splits into RM + WIP + FG holding periods + debtors – creditors.

Worked mini-example (reverse ratios)

Current Ratio 2.5:1, Quick 1.5:1, WC ₹6,00,000, Inventory TO 8× on COGS, GP 25%, collection 36.5 days (365-yr).

- CL: $2.5x - x = 1.5x = 6,00,000 \rightarrow x = \text{CL ₹4,00,000, CA ₹10,00,000}$.
- Quick Assets = $1.5 \times 4,00,000 = 6,00,000 \rightarrow \text{Inventory} = 10,00,000 - 6,00,000 = \text{₹4,00,000}$.
- COGS = $8 \times 4,00,000 = 32,00,000$; Sales = $32,00,000 \div 0.75 = \text{₹42,66,667}$.
- Debtors = $36.5 \times 42,66,667 \div 365 = \text{₹4,26,667}$ (fits inside quick assets ✓).

Exam traps & must-remember

1. Quick assets also remove **prepaid expenses**, not just inventory.
2. Bank overdraft in quick ratio — chronic OD sometimes excluded; **state assumption**.
3. Debtors turnover uses **credit sales**; creditors uses **credit purchases**.
4. Turnover ratios use **average** (opening+closing)/2 when opening available.
5. Capital Employed two routes must reconcile (TA – CL = SHF + LTD); else misclassification.
6. ROCE numerator = **EBIT** (pre-interest, pre-tax), not PAT.

7. EPS = earnings **after** preference dividend ÷ equity shares.
8. High ratio ≠ good (CR 6:1 = idle cash; high debtors period = lax credit).
9. Slower payables = often favourable (free finance); not "faster = better."
10. DuPont ROE denominator must be consistent (total vs equity-only funds).
11. Inventory turnover = **COGS ÷ Inventory** (both at cost), not Sales.
12. Fictitious assets (preliminary exp, discount on shares, accumulated losses) → deduct from SHF & total assets.
13. Gross up post-tax charges (pref div, principal) by ÷(1-t) in coverage ratios.
14. Use correct units: "2:1", "6 times", "60 days", "%".
15. Effect-on-ratio: ratio >1 → equal addition to both sides pulls down toward 1, equal subtraction pushes up; <1 reverses; intra-side swap = no change.
16. Times ↔ days are reciprocals scaled by 360/365.

One-line recall

- Ratio manufactures a yardstick; judge against trend, peer, industry — never in isolation.
- Four families: Liquidity, Solvency, Activity, Profitability/Valuation.
- Capital Employed = TA – CL = SHF + LTD; Op Ratio + Op Profit Ratio = 100%; Debt Ratio + Proprietary = 100%.
- DuPont ROE = margin × turnover × leverage (must reconcile to direct PAT/SHF).
- SGR = ROE × retention; Earnings Yield = 1/(P/E).
- Window-dressing: paying a creditor before B/S date raises current ratio (>1) without real improvement.

Cost of Capital

Snapshot

Cost of capital = minimum return a project must earn to leave firm value unchanged = the **hurdle rate**. WACC (K_o) is the blended cost of the whole capital pool; projects must beat it to create value. Build WACC bottom-up: each component cost, then weighted blend.

Core concepts

- **Three equivalent definitions:** financing view (return demanded by suppliers), opportunity-cost view (next-best return of equal risk), break-even view (return keeping share price unchanged).
- **Component decomposition:** $K = r_o$ (riskless) + b (business-risk premium) + f (financial-risk premium). Equity carries both premiums fully → dearest.
- **Explicit cost** (cash paid: interest/dividend) vs **implicit cost** (opportunity, e.g. retained earnings: zero explicit, real implicit).
- **Ordering (self-check):** $K_d < K_p < K_r \leq K_e$. Interest is tax-deductible; dividends are not → compute K_d post-tax.
- **Pooling principle:** use pooled target-structure WACC, not the specific cheap source funding a project (financing is fungible).
- **Average** (existing pool) vs **marginal** (next rupee raised) — new projects use marginal.

Key provisions / rules

Source	Formula	Notes
Irredeemable debt	$K_d = I(1-t) \div NP$	tax shield on interest only
Redeemable debt	$K_d = [I(1-t) + (RV-NP)/n] \div [(RV+NP)/2]$	premium term NOT tax-adjusted
Debt by YTM	$K_d = L + [NPV_L \div (NPV_L - NPV_H)] \times (H - L)$	exact/interpolation when demanded
Term loan	$K_d = \text{rate} \times (1-t)$	no premium/discount
Zero-coupon bond	$K_d = (RV/NP)^{(1/n)} - 1$ (pre-tax)	discount accretes; tax per problem
Irredeemable preference	$K_p = PD \div NP$	no $(1-t)$
Redeemable preference	$K_p = [PD + (RV-NP)/n] \div [(RV+NP)/2]$	no $(1-t)$
Equity — Gordon	$K_e = D_1/P_0 + g$	$D_1 = D_0(1+g)$; use NP for fresh issue
Equity — CAPM	$K_e = R_f + \beta(R_m - R_f)$	only systematic risk priced
Equity — Bond yield + premium	$K_e = \text{firm's debt yield} + \text{risk premium}$	fallback
Equity — Earnings yield	$K_e \approx E/P$	only under zero growth + full payout
Retained earnings	$K_r = K_e$	opportunity cost; NOT free
Retained (with adj.)	$K_r = K_e(1-tp)(1-b)$	only if personal tax/brokerage given
WACC	$K_o = \sum (\text{component cost} \times \text{weight})$	all costs post-tax
Break point	$(\text{Amount of cheaper source}) \div (\text{weight of that source})$	—

- **Growth rate:** $g = (D_{\text{now}}/D_{\text{past}})^{(1/n)} - 1$ (count *intervals*, not data points); or $g = b \times r$ (retention $\times$ ROE).
- **Weights hierarchy (for financing decision):** **marginal/target** > **market** > **book**. Market weights preferred because they reflect today's opportunity cost; book weights are historical/sunk. Retained earnings folded into equity's market value.
- **Post-tax throughout** — WACC discounts post-tax cash flows; never mix pre-tax K_d or market weight with book cost.
- **Tax-shield caveat:** worthless if firm has no taxable profit → use pre-tax coupon.
- **Marginal cost of capital (MCC):** WACC of incremental funds; build tranche-by-tranche across break points (can rise OR fall). Pair MCC schedule with Investment Opportunity Schedule → optimal capital budget at the crossing.
- **Sanity test:** WACC lies between lowest and highest component cost.

Worked mini-example

Fresh-capital WACC with retained-earnings break point. Target: Equity 50%, Pref 10%, Debt 40%; tax 30%. Retained earnings ₹15,00,000. $K_e(\text{retained})=18.00\%$, $K_e(\text{fresh})=18.71\%$, $K_p=12.04\%$, $K_d=7.69\%$. Raise ₹50,00,000.

- Break point = $15,00,000 \div 0.50 = \text{₹}30,00,000$.
- MCC first ₹30L (retained equity): $0.50 \times 18.00 + 0.10 \times 12.04 + 0.40 \times 7.69 = \mathbf{13.28\%}$.
- MCC beyond ₹30L (fresh equity): $0.50 \times 18.71 + 0.10 \times 12.04 + 0.40 \times 7.69 = \mathbf{13.64\%}$.
- Weighted for full ₹50L: $(30 \times 13.28 + 20 \times 13.64) / 50 = \mathbf{13.42\%} \rightarrow$ hurdle for the raise; a 13% project is rejected.

Exam traps & must-remember

1. Forgetting tax shield on debt — only **interest** gets it, never preference/equity dividends.
2. Do NOT tax-adjust the redemption premium/discount term $(RV-NP)/n$.
3. Retained earnings NOT free $\rightarrow K_r = K_e$.
4. Use $D_1 (= D_0(1+g))$, not D_0 , in Gordon.
5. Fresh issues use **net proceeds** (net of flotation), not market price $\rightarrow$ fresh equity > retained.
6. Book vs market weights: use market if given; fold retained earnings into equity market value.
7. $K_e < K_d$ is impossible — signals an error.
8. New project on new money $\rightarrow$ hurdle is **marginal** cost, not historical WACC.
9. CAPM premium = $\beta(R_m - R_f)$ added to R_f ; not $\beta \times R_m$.
10. Weights must sum to 1.0000 (show totals row).
11. Miscounting growth period n (year 0 to 5 = 5 intervals).
12. Never mix pre-tax cost with market weight, or book weight with market cost.
13. MCC does not only rise — a later cheaper tranche can lower a step.
14. Loss-making firm $\rightarrow$ no tax shield $\rightarrow$ use pre-tax coupon.
15. DDT abolished post-2020 — only gross up preference dividend if problem says so.
16. Earnings yield $E/P = K_e$ only at zero growth + full payout.

One-line recall

- Cost of capital = hurdle rate; beat WACC/MCC $\rightarrow$ create value, else destroy.
- $K_d < K_p < K_r \leq K_e$; interest tax-deductible so $K_d = I(1-t)$.
- Redeemable: $[\text{annual cost}] \div [\text{average funds}]$; premium term not tax-adjusted.
- Gordon $K_e = D_1/P_0 + g$; CAPM $K_e = R_f + \beta(R_m - R_f)$; $K_r = K_e$ (opportunity cost, not free).
- Weights: marginal > market > book; all costs post-tax; WACC lies between extremes.
- Break point = cheap source $\div$ its weight; build MCC schedule tranche-by-tranche.

Capital Structure & Leverage

Snapshot

The financing decision trades higher expected owner return (cheap, tax-deductible debt) against higher risk of ruin (fixed interest paid in bad years too). **Leverage** measures how cost/financing structure magnifies profit swings; **capital structure theory** diagnoses whether adding debt makes owners richer or just bumpier. Value is highest exactly when WACC is lowest (Value = operating cash flow ÷ WACC).

Core concepts

- **Vocabulary guard:** Capital structure = long-term mix only. Financial structure = whole liabilities side. Capitalisation = total long-term amount. Over-capitalised (too much total capital for earnings, low ROI) ≠ over-gearred (too much debt) ≠ over-trading (too little capital for sales).
- **Operating leverage** (fixed operating costs; sales→EBIT) → **business risk** (from asset/technology choice). **Financial leverage** (fixed interest; EBIT→EPS) → **financial risk** (from financing choice). Business risk exists before financing (EBIT computed pre-interest).
- Leverage is highest **near its break-even** and decays toward 1 as you move away; always "at a stated activity level," never a fixed firm property.
- **maximise value** ⇔ **minimise WACC** — the axis of all four theories.

Key provisions / rules

Income skeleton: Sales – Variable Cost = **Contribution** – Fixed Cost = **EBIT** – Interest = **EBT** – Tax = **PAT** – Pref Div = Equity earnings ÷ N = **EPS**.

Concept	Formula	Notes
DOL	Contribution ÷ EBIT	Sales→EBIT; =1 if no fixed cost
DOL (units)	$Q \div (Q - BEQ)$	explodes near break-even
DFL	$EBIT \div (EBIT - \text{Interest}) = EBIT/EBT$	EBIT→EPS; =1 if no debt
DFL (with pref.)	$EBIT \div [EBIT - I - PD/(1-t)]$	gross up preference dividend
DCL	$DOL \times DFL = \text{Contribution} \div EBT$	Sales→EPS; total risk
EPS	$[(EBIT - I)(1-t) - PD] \div N$	N = existing + new shares
Financial break-even EBIT	$I + PD/(1-t)$	EBIT where EPS = 0
Operating break-even (units)	$\text{Fixed cost} \div (s - v)$	EBIT = 0
Indifference EBIT	solve $EPS_1 = EPS_2$	above → pick more debt
Favourable leverage test	$ROI (\text{EBIT}/\text{capital}) > K_d$	else debt drags EPS down

Reverse-engineering: $DCL \div DOL = DFL$. Contribution = $DOL \times EBIT$; Fixed cost = Contribution – EBIT; $EBT = EBIT/DFL$; Interest = $EBIT - EBT$; Debt = Interest ÷ rate. "Financial leverage = 2" usually means DFL (elasticity); "highly geared" means D/E amount.

Capital structure theories ($V = S + D$; $K_d < K_e$ always):

Feature	NI	NOI	Traditional	MM no-tax	MM tax
Held constant	K_e & K_d	K_o	neither	K_o	—
K_e as debt rises	constant	rises	rises (mild→sharp)	rises linearly	rises
WACC as debt rises	falls	constant	U-shaped	constant	falls
Optimal structure	100% debt	none	sweet spot	none	≈100% debt
Value	$S = (EBIT - I)/K_e$; $V = S + D$; $K_o = EBIT/V$	$V = EBIT/K_o$	—	$V_L = V_U$	$V_L = V_U + tD$

- **NI:** K_e, K_d constant $\rightarrow$ WACC falls, gear to 100%.
- **NOI:** K_o fixed by business risk; $V = EBIT/K_o$; $K_e = K_o + (K_o - K_d)(D/S)$; structure irrelevant.
- **Traditional:** U-shaped WACC, three stages (low gearing WACC falls $\rightarrow$ optimal zone $\rightarrow$ high gearing both K_e & K_d climb, WACC rises); optimum at bottom.
- **MM no-tax:** $V_L = V_U$ by arbitrage/homemade leverage; Prop II: $K_e = K_o + (K_o - K_d)(D/S)$.
- **MM tax:** $V_L = V_U + t \times D$; $V_U = EBIT(1-t)/K_e$; shield = tD (K_d cancels, assumes permanent debt + enough profit).
- **Trade-off theory:** $V_L = V_U + PV(\text{tax shield}) - PV(\text{financial distress \& bankruptcy costs})$. Distress: *direct* (legal/admin fees) vs *indirect* (lost customers/suppliers/staff, forgone projects — usually larger). Optimum where marginal tax benefit = marginal distress cost.
- **Pecking order (Myers):** retained earnings $\rightarrow$ debt $\rightarrow$ equity last (info asymmetry); no target ratio.

Common theory assumptions: only debt & equity; EBIT/assets fixed; 100% payout; constant business risk; rational investors (MM: perfect markets, no tax base case, equal borrowing rate).

Worked mini-example (all three leverages)

50,000 units $\times$ ₹40; VC ₹24/unit; fixed ₹4,00,000; 10% debentures ₹10,00,000; tax 30%.

- Contribution = $50,000 \times 16 = 8,00,000$; EBIT = $8,00,000 - 4,00,000 = 4,00,000$; Interest = 1,00,000; EBT = 3,00,000.
- DOL = $8,00,000 / 4,00,000 = 2.0$; DFL = $4,00,000 / 3,00,000 = 1.33$; DCL = $2.0 \times 1.33 = 2.67$ ($= 8,00,000 / 3,00,000$).
- Break-even = $4,00,000 / 16 = 25,000$ units; firm at $2 \times$ BEQ $\rightarrow$ DOL = $50,000 / (50,000 - 25,000) = 2.0 \checkmark$. At 30,000 units DOL leaps to 6.0.

Exam traps & must-remember

1. Preference dividend must be **grossed up** by $/(1-t)$ in DFL and financial break-even.
2. "More leverage is good" $\neq$ "safe" — below indifference/break-even EBIT, high leverage gives *lower*/negative EPS.
3. DOL, DFL are not constants — computed at a base level; explode near break-even.
4. DFL = EBIT/EBT only valid with **no** preference shares.
5. **NI vs NOI:** NI capitalises income-to-equity (residual, optimum trend); NOI capitalises whole-firm operating income (irrelevant).
6. MM no-tax $V_L = V_U$; MM tax $V_L = V_U + tD$; state which world.
7. Homemade leverage: investor **borrowes personally** to buy the undervalued unlevered firm (unwinds/lends if levered firm cheaper) — know both directions.
8. Include existing + newly issued shares in EBIT-EPS.
9. DCL = Contribution/EBT (not /EBIT = DOL, not EBIT/EBT = DFL).
10. Leverage magnifies both directions; falling sales $\rightarrow$ same multiplier negative.
11. Trading on equity benefits owners only when **ROI > K_d** .
12. MM-tax: capitalise EBIT $(1-t)$ at K_e for V_U (not pre-tax EBIT); shield = tD regardless of rate.
13. Over-capitalisation $\neq$ over-gearing $\neq$ over-trading.
14. Valuation tables use **market** values (WACC = EBIT/V on market values), not book unless stated.

One-line recall

- DOL = Contribution/EBIT; DFL = EBIT/EBT (gross up pref div); DCL = DOL $\times$ DFL = Contribution/EBT.
- Financial break-even EBIT = $I + PD / (1-t)$; EPS = $[(EBIT - I)(1-t) - PD] / N$.
- Above indifference EBIT $\rightarrow$ more debt; below $\rightarrow$ more equity; sanity-check ROI > K_d .
- NI: WACC falls (100% debt). NOI/MM-no-tax: WACC flat (irrelevant). Traditional: U-shape (optimum). MM-tax: $V_L = V_U + tD$.
- Trade-off = tax shield vs distress cost; Pecking order = internal $\rightarrow$ debt $\rightarrow$ equity.
- Leverage magnifies feast and famine, largest near break-even.

Capital Budgeting (Investment Decisions)

Snapshot

Deciding which long-term assets to buy. Inputs = incremental, after-tax, cash flows (not profit). Yardstick = NPV (wealth in ₹). Decision categories dictate the technique:

- **Accept–reject (independent):** accept every $NPV > 0$ ($\Leftrightarrow IRR > k \Leftrightarrow PI > 1$). NPV & IRR never conflict.
- **Mutually exclusive:** pick one; NPV vs IRR can disagree (scale/timing).
- **Capital rationing:** budget capped; select combination maximising total NPV (PI's home).
- **Replacement / abandonment:** incremental analysis; economic life may be < physical life.

Examiner signals the box in the first 2 lines ("only one of..." = mutually exclusive; "funds limited to ₹X" = rationing).

Core concepts

- **Cash not profit; incremental not total; after-tax; post opportunity cost.** Goal = maximise shareholder wealth.
- **k = cost of capital (WACC)** = opportunity cost = reinvestment assumption. NPV reinvests interim flows at k (neutral, defensible); IRR reinvests at IRR (often unrealistic) — the deep reason NPV beats IRR.
- **Five cash-flow rules:** (1) incremental (with-minus-without); (2) ignore sunk costs; (3) include opportunity costs; (4) include working capital (out at start/on rise, recovered at end, untaxed, not depreciated); (5) ignore financing flows (interest/dividend) in the cash line — they live in k (else double-count).
- **Depreciation:** non-cash; enters only via tax. Add back after computing tax.
- **Cannibalisation/erosion:** lost contribution on old product = incremental outflow. Complementary lift = inflow.

Key provisions / rules

After-tax operating cash flow build-up

Line	Item
PBDT	Sales – cash operating costs
PBT	PBDT – Depreciation
Tax	$PBT \times t$
PAT	$PBT - \text{Tax}$
CF	$PAT + \text{Depreciation}$

Item	Formula
After-tax operating CF	$PBDT \times (1 - t) + \text{Depreciation} \times t$
Depreciation tax shield	$\text{Depreciation} \times t$
SLM depreciation	$(\text{Cost} - \text{Salvage}) \div \text{Life}$
WDV depreciation	Rate $\times$ opening WDV each year (front-loaded shields $\rightarrow$ higher PV)
Initial outlay	Asset + Installation + Initial WC + Opportunity cost
Terminal inflow	Salvage (net of tax) + WC recovered

Tax on salvage (4 cases)

Case	Treatment
Salvage = book value	No tax; salvage enters gross
Salvage > book value	Profit = Salvage – BV; net salvage = Salvage – $t \times (\text{Salvage} - \text{BV})$
Salvage < book value	Loss saves tax; net salvage = Salvage + $t \times (\text{BV} - \text{Salvage})$

Case	Treatment
Salvage = 0, BV > 0	Terminal loss = BV; tax saving = BV × t

Techniques

Technique	Formula	Rule	Time value
Payback (even)	Initial Investment ÷ Annual CF	≤ target	No
Payback (uneven)	Completed yrs + (Unrecovered ÷ next-yr CF)	≤ target	No
Discounted Payback	Accumulate PV of CFs till outlay recovered	≤ target	Yes
ARR	Avg Annual PAT ÷ Avg Investment × 100	≥ target	No
NPV	$\sum CF_t \div (1+k)^t - \text{Investment}$	> 0; highest	Yes
IRR	Rate where NPV = 0	> k	Yes
MIRR	$(\text{Terminal Value of inflows} \div \text{PV of outflows})^{(1/n)} - 1$	> k	Yes
PI	$\text{PV of inflows} \div \text{Investment} = 1 + \text{NPV} / \text{Investment}$	> 1	Yes
EAC	Total PV of costs ÷ PVIFA(k, life)	Lowest EAC wins	Yes

- **Avg Investment (ARR)** = (Cost – Salvage) ÷ 2 + Salvage + Additional WC. (Salvage & WC at full value; depreciable part halved.) ICAI default = average investment (state basis).
- **IRR interpolation:** $\text{IRR} = L + [\text{NPV}_L \div (\text{NPV}_L - \text{NPV}_H)] \times (H - L)$. Bracket L,H tightly (2–3% apart) around zero.
- **Payback reciprocal** ($1 \div \text{payback}$) ≈ IRR for long, even-cash-flow projects.
- **Discounted payback within life** ⇒ **NPV > 0** (sufficient condition).
- **PI for rationing:** valid only if projects divisible & single-period. Indivisible → test feasible combinations for max total NPV.

Special decisions

- **Replacement:** Year-0 = New cost – after-tax proceeds of old asset (add tax saving if sold below BV; deduct tax if above). Operating = incremental CF; **incremental depreciation = new dep – old dep forgone** (drives shield). Terminal = incremental salvage. (The two "old-asset-forgone" legs earn/lose marks.)
- **Unequal lives (mutually exclusive):** EAC (NPV ÷ PVIFA of own life; lowest cost / highest benefit wins) or replacement-chain over LCM of lives. EAC preferred.
- **Abandonment:** find optimal year — compare "value if abandoned now" vs "PV of (next year CF + next year salvage)".

NPV vs IRR conflict — only for mutually exclusive. Roots: **scale** and **timing** (reinvestment rate). Resolve: trust NPV. Confirm via **incremental IRR** = IRR of (Large – Small) CFs; if > k, bigger-NPV project wins. **Fisher crossover rate** = incremental IRR = rate where the two NPVs are equal. **Multiple/no IRR:** non-conventional flows (>1 sign change) → use NPV or MIRR.

Worked mini-example

Plant ₹20,00,000 + WC ₹3,00,000 = ₹23,00,000 outlay. Life 4 yr, salvage ₹2,00,000, SLM, t = 30%, k = 12%.

- Depreciable base = 20,00,000 – 2,00,000 = 18,00,000; dep = ₹4,50,000/yr.
- Operating CFs (PBDT×0.70 + 4,50,000×0.30): Yr1 6,25,000; Yr2 7,65,000; Yr3 8,35,000; Yr4 6,60,000.
- Terminal Yr4: salvage 2,00,000 (= BV → no tax) + WC 3,00,000 = 5,00,000. Yr4 total = 11,60,000.
- PV @12%: 5,58,125 + 6,09,705 + 5,94,520 + 7,37,760 = 25,00,110.
- **NPV = 25,00,110 – 23,00,000 = ₹2,00,110 > 0 → Accept.**

Variation A: salvage ₹3,50,000 > BV 2,00,000 → profit 1,50,000, tax 45,000, net salvage 3,05,000 (depreciation schedule unchanged — still based on expected ₹2,00,000).

Exam traps & must-remember

1. Deducting a **sunk cost** (past feasibility study) — ignore it.
2. Forgetting **opportunity cost** of owned land/building/old asset that could be sold.
3. **Subtracting interest** from cash flows — never; it's in k .
4. Treating **depreciation as cash outflow** — only tax effect is cash.
5. Omitting **WC recovery** in final year; or depreciating/taxing WC.
6. **Tax on salvage** only if salvage $\neq$ book value. Machine depreciated to nil but sold for scrap = taxable profit.
7. Using **PAT instead of CF** (forgetting add-back of depreciation).
8. Ranking mutually exclusive by **IRR/PI** (scale illusion) — NPV decides; confirm with incremental IRR.
9. **Multiple/no IRR** on non-conventional flows — use NPV/MIRR, don't interpolate blindly.
10. Mixing **nominal/real** — ICAI problems are nominal.
11. **ARR base** — average vs initial; default average, state it.
12. **Wrong depreciable base** — SLM on (Cost – Salvage), not Cost.
13. Comparing **unequal-life** projects on raw NPV — use EAC.
14. Replacement — dropping the **old-asset-forgone** legs (after-tax sale + incremental depreciation).
15. **Incremental WC** when WC = % of rising sales — outflow only the year-on-year increase; recover accumulated balance at end.
16. **Interpolating IRR** over too wide a bracket.
17. **Loss-year tax** = saving only if firm has other taxable profits; else deferred.
18. **WDV terminal balancing charge/allowance** almost always arises (BV $\neq$ salvage) — tax it.

One-line recall

- Cash not profit; incremental not total; sunk gone, opportunity counts.
- Depreciation only for tax, then add it back; WC out at start, back at end, untaxed.
- Tax the salvage only if it differs from book value; SLM base = Cost – Salvage.
- NPV rules; when in doubt, ask "am I richer today?" IRR reinvests at IRR (flatters); NPV reinvests at k (honest).
- Unequal lives $\rightarrow$ annualise (EAC); replacement $\rightarrow$ incremental (incl. incremental depreciation).
- Rationing: divisible $\rightarrow$ rank PI; indivisible $\rightarrow$ best feasible combination by total NPV.

Risk Analysis in Capital Budgeting

Snapshot

A single-point NPV is a conditional statement ("if inputs hold, value is created") — it says nothing about likelihood, spread, or which assumption is fragile. Two honest responses to risk:

- **Charge for risk** (change the accept/reject rule): RADR, Certainty Equivalent.
- **Describe the spread** (understand outcomes): sensitivity, scenario, expected NPV + σ + CV, decision trees, simulation.

These are not rivals — a full appraisal does both. The question's data structure chooses the rung for you.

Core concepts

- **Risk vs Uncertainty (Knight):** Risk = probabilities knowable (dice); Uncertainty = probabilities not reliably attachable. Sensitivity/scenario need no probabilities; expected NPV/simulation require them.
- **Probability sources:** objective (historical frequency), subjective (judgement), blend. The moment a question gives a probability, it has moved you into "risk" → full quantitative toolkit available.
- **Diversifiable vs non-diversifiable:** unsystematic/unique (machine breakdown, fire) diversifies away; systematic/market (GDP, interest, inflation) does not. Under CAPM a diversified owner is rewarded only for systematic risk. σ = total risk; β = systematic only.

Key provisions / rules

Concept	Formula
RADR	$RADR = R_f + \text{Risk premium}; NPV = \sum CF_t \div (1+RADR)^t - CF_0$
Certainty Equivalent	$NPV = \sum (\alpha_t \times CF_t) \div (1+R_f)^t - CF_0$
CE coefficient	$\alpha_t = \text{Certain CF} \div \text{Expected risky CF} (0 \leq \alpha \leq 1)$
CE–RADR reconciliation	$\alpha_t = (1+R_f)^t \div (1+RADR)^t$
RADR from α (year t)	$RADR_t = (1+R_f) \div \alpha_t^{1/t} - 1$
Expected cash flow	$\bar{CF} = \sum p_i \times CF_i$
Expected NPV	$\sum \bar{CF}_t \div (1+r)^t - CF_0$
Standard deviation	$\sigma = \sqrt{\sum p_i (CF_i - \bar{CF})^2}$
Coefficient of variation	$CV = \sigma \div \text{Expected value}$
σ of NPV (independent)	$\sqrt{\sum \sigma_t^2 \div (1+r)^{2t}}$
σ of NPV (perfectly correlated)	$\sum \sigma_t \div (1+r)^t$
Sensitivity margin (%)	$NPV \div \text{PV of the variable} \times 100$
Joint prob (tree terminal node)	product of branch probabilities along the path

RADR: higher risk → higher premium → higher hurdle → lower NPV. Adjust *around* WACC (riskier project above WACC, safer below). Premium sources: managerial judgement, risk-class buckets, or CAPM [$R_f + \beta(R_m - R_f)$]. **Flaw:** forces risk to compound at a constant rate every year — over-penalises long/back-loaded projects. WACC-for-everything trap: over-accepts risky, wrongly rejects safe projects.

CE: separates time value (discount at R_f) from risk (α_t in numerator, per year). Theoretically superior (each year its own α). **Lower α = more risk.** RADR *implicitly* assumes α falls with t (reconciliation formula). A **rising α** (risk falling over time — start-up) cannot be reproduced by any positive constant RADR → CE only honest tool. Constant RADR matches stated α 's only if they decline in the exact geometric pattern.

Feature	RADR	CE
What is adjusted	Discount rate (denominator)	Cash flows (numerator)
Rate used	Risk-free + premium	Risk-free only
Risk each year	Grows at compound rate	Set independently per year
Ease	Easy, board-friendly	Harder, subjective
Soundness	Weaker (mixes risk & time)	Stronger (separates)

Sensitivity analysis: flex one input $\pm X\%$, hold rest constant; biggest NPV swing / thinnest break-even margin = most critical (guard it). Needs no probabilities. Denominator of margin = **whole-life PV of that variable's stream** (not one year's figure). Limits: moves variables in isolation (unrealistic — recession moves all together); tells what is sensitive, not how likely.

Scenario analysis: internally consistent whole states (pessimistic / most likely / optimistic), every variable set together — captures correlation. Attach probabilities → becomes a coarse 3-point expected NPV.

Expected NPV, σ , CV: σ = absolute risk in ₹ (same units as CF, being variance). **CV = risk per rupee of return** — correct comparator when sizes/means differ; lower CV better. σ ranks only when means equal. CV meaningless if mean near zero/negative — often compute CV on expected *cash flow* (safely positive); state base. **Independent** cash flows → variances add (discount factor squared); **perfectly correlated** → std devs add (plain factor). Correlated gives larger σ (no year-to-year cancelling).

Decision trees: decision nodes (squares, you choose), chance nodes (circles, nature decides). **Fold back right→left:** expected value at chance nodes, **maximum** at decision nodes. Replace a negative continuation branch with **₹0 (abandon)**. Multiply conditional probabilities along the branch (joint prob); terminal joint probs sum to 1. Discount future flows to PV before folding if raw flows + rate given. Captures value of flexibility (seed of real options).

Simulation (Monte Carlo, Hertz 1964): assign a probability distribution to each input, sample repeatedly (draw → compute NPV → repeat thousands), plot NPV distribution. Gives mean, σ , and **P(NPV < 0)**. Handles many correlated variables. Data-hungry, model-dependent; does not itself make the decision.

Worked mini-example

CE vs RADR. Outlay ₹6,00,000; CF ₹3,00,000/yr for 3 yrs; $\alpha = 0.90, 0.80, 0.70$; $R_f = 6\%$, RADR = 12%.

- **CE:** certain CFs 2,70,000 / 2,40,000 / 2,10,000 $\times$ PVF@6% (0.9434/0.8900/0.8396) = 2,54,718 + 2,13,600 + 1,76,316 = 6,44,634. NPV = **₹44,634**.
- **RADR:** 3,00,000 $\times$ PVF@12% (0.8929/0.7972/0.7118) = 7,20,570. NPV = **₹1,20,570**.
- **Why differ:** RADR-implied $\alpha = (1.06/1.12)^t = 0.946 / 0.896 / 0.848$ — gentler than management's stated 0.90/0.80/0.70. Management more cautious than 12% reflects → lower CE NPV. Methods reconcile only when stated α = RADR-implied α .

Exam traps & must-remember

1. **σ vs CV:** equal means → rank by σ ; different means/sizes → must use CV.
2. **Wrong σ _NPV formula:** "independent" → $\sqrt{\sum \sigma_t^2 / (1+r)^{2t}}$ (square the factor); "perfectly correlated" → $\sum \sigma_t / (1+r)^t$ (plain). ~73% gap.
3. **CE discount rate = risk-free**, never RADR (double-counts risk).
4. **RADR direction:** higher risk → higher rate → lower NPV. Premium is *added*.
5. **α direction:** lower α = more risk; rising α impossible under any positive constant RADR.
6. Sensitivity finds the **most dangerous to mis-estimate**, not "the good" variable.
7. **Tree fold-back right→left:** expected value at circles, **maximum** at squares (you control decisions — don't average).
8. Negative continuation branch → replace with **₹0 (walk away)**, not the negative number.
9. Probabilities sum to 1; joint prob of terminal node = product along path.
10. Expected NPV may be a value that never actually occurs — it's a long-run average.

11. **Salvage under RADR** is discounted at the RADR (only CE discounts anything at R_f).
12. Sensitivity margin denominator = whole-life PV of the variable's stream.
13. σ is in rupees ($\sqrt{\text{variance}}$); variance is ₹^2 — don't quote it as "the risk."
14. WACC-for-everything ignores that riskier projects need a higher hurdle.

One-line recall

- RADR punishes the rate; CE punishes the cash flow. Only CE discounts at the risk-free rate.
- σ is risk in rupees; CV is risk per rupee. Equal means $\rightarrow \sigma$; different means $\rightarrow \text{CV}$.
- Independent years cancel (variances add, factor squared); correlated years compound (σ 's add).
- Lower α = more risk; a rising α needs CE (no constant RADR can make it).
- Tree: fold back right $\rightarrow$ left, max at squares, expected at circles, ₹0 for abandon.
- Simulation = scenario analysis with scenarios turned up to infinity; gives $P(\text{NPV} < 0)$.

Dividend Decisions

Snapshot

How much profit to pay out vs retain. The intellectual spine: **does the payout split change share value, or is it irrelevant?** Relevance school (Walter, Gordon, Graham–Dodd) says it matters; Irrelevance school (MM) proves it doesn't in perfect markets. All hinge on **r vs Ke** and real-world frictions (taxes, issue costs, information gaps).

- **Wealth = market price + dividend received** (not price alone).
- **Dividend policy** (long-run rule) $\neq$ single **dividend decision** (this year's declaration). Theories evaluate the policy.

Core concepts

Notation: E = EPS; D = DPS; r = return on retained earnings; Ke = cost of equity (required); P = price; b = retention ratio = $(E-D)/E$; $(1-b)$ = payout = D/E ; g = growth = $b \times r$.

Identity chain: payout = $D/E = 1-b$; $b = (E-D)/E$; $g = br$; $b = g/r$. Fix decimals vs % at top and convert before substituting.

The one comparison ruling relevance:

Firm	Relationship	Optimal payout
Growth	$r > Ke$	0% (retain all)
Declining	$r < Ke$	100% (distribute all)
Normal	$r = Ke$	Indifferent

Key provisions / rules

Model	Core formula	School / lever
Walter (1963)	$P = [D + (r/Ke)(E - D)] \div Ke$	Relevance; r vs Ke
Gordon (1962)	$P = E(1 - b) \div (Ke - br) = D \div (Ke - g)$	Relevance; bird-in-hand
Gordon reversed	$Ke = D_1/P + g = D_1/P + br$	Cost of equity
MM (1961)	$P_0 = (P_1 + D_1) \div (1 + Ke)$	Irrelevance; homemade dividends
MM firm value	$nP_0 = [(n + \Delta n)P_1 - I + E] \div (1 + Ke)$	
MM new shares	$\Delta n \times P_1 = I - (E - nD_1)$	
Graham–Dodd	$P = m(D + E/3)$	Strong relevance (dividends $\sim 3 \times$ retentions)
Linter	$D_1 = D_0 + [(target\ payout \times EPS) - D_0] \times adjustment\ factor$	Sticky dividends

Walter: always a corner solution (0% or 100%) — coefficient on D is $(1 - r/Ke)/Ke$: negative if $r > Ke$ (push D to 0), positive if $r < Ke$ (push D to 100%). Assumptions: all-internal finance, constant r & Ke, constant EPS/DPS, infinite life. Criticism: constant r unreal (marginal r falls); entangles investment & dividend decisions.

Gordon: $g = br$ (growth only from ploughback earning r). Guardrail **Ke > br** (else denominator ≤ 0 , price infinite). Bird-in-hand: risk-averse investors value certain near dividends over uncertain gains; distant dividends discounted higher. Reverse form: grow $D_0 \rightarrow D_1 = D_0(1+g)$ when dividend "just paid." Assumptions: all-equity, no external finance, constant r & Ke, $Ke > br$, constant b, tax ignored.

MM: homemade dividends (investor sells/buys shares costlessly to make own cash pattern) $\rightarrow$ arbitrage $\rightarrow$ dividend irrelevant. D_1 **cancels** when Δn substituted — that cancellation *is* the proof. Bigger dividend $\rightarrow$ smaller retained earnings $\rightarrow$ bigger new issue $\rightarrow$ new shareholders capture exactly what old ones took as dividend (transfer, not value). If retained earnings $\geq I$, $\Delta n \leq 0$ (buyback, no fresh issue). New shares raised at P_1 (year-end), not P_0 . Assumptions: perfect markets, no taxes, no flotation costs, no transaction costs, fixed investment policy, perfect certainty.

How real world revives relevance: taxes differ (tax-preference for retention/buyback), flotation costs real (retention cheaper), transaction costs real (cash dividend serves income-seekers), information asymmetric (dividend signals).

Residual theory: dividend = earnings left after funding all $r > K_e$ projects from (cheapest) retained earnings. Explains why growth firms pay nil, mature firms pay heavily. Implies volatile dividends (clashes with Linter's smoothing). Not a pricing formula.

Forms of dividend: cash; **stock dividend/bonus** (capitalises reserves → raises paid-up share capital, face value unchanged, reserves fall; wealth unchanged); **stock split** (only re-denominates face value ₹10 → 2×₹5, no reserve capitalised, total capital unchanged); scrip/bond. Bonus vs split: bonus increases paid-up capital, split does not.

Buyback (Sec 68–70): alternative route to return surplus cash; share count falls → EPS rises; tax-efficient (capital-gains route, verify current tax); flexible for one-off surplus (protects sticky-dividend signal); signals undervaluation, supports control. Dividend & buyback = two taps on same cash tank. EPS rise ≠ wealth creation (concentration only).

Legal (Companies Act 2013 — verify current text): Sec 123 — dividend only out of (a) current profits after depreciation, (b) accumulated past profits, or (c) government money; interim from surplus; transfer to reserves now optional. Unpaid dividend → Unpaid Dividend A/c → IEPF after 7 years.

Factors (L-E-G-A-L + C): Liquidity, Earnings stability, Growth/financing needs, Access to capital, Legal & contractual limits + Clientele, Control, Confidence-signalling; also taxation, inflation. Structure answer as internal / shareholder-facing / external.

Worked mini-example

Walter, $E=₹10$, $K_e=12\%$, growth firm $r=15\%$ ($r/K_e=1.25$):

Payout	D	$(r/K_e)(E-D)$	Numerator	$P = \div 0.12$
0%	0	12.50	12.50	₹104.17
50%	5	6.25	11.25	₹93.75
100%	10	0	10.00	₹83.33

Max at 0% payout → retain all. (Normal firm $r=K_e$: $P = E/K_e = ₹83.33$ at any payout.)

Reverse Gordon: $P=₹120$, $D_0=₹6$, $b=40\%$, $r=20\%$. $g = 0.40 \times 0.20 = 8\%$. $D_1 = 6 \times 1.08 = ₹6.48$ (grow it!). $K_e = 6.48/120 + 0.08 = 0.054 + 0.08 = 13.4\%$. (Using D_0 gives wrong 13%.)

Exam traps & must-remember

1. Confusing r (firm earns) and K_e (shareholders require) — inverts every conclusion.
2. Percent vs decimal in r/K_e : $15\%/12\% = 1.25$, not $0.15/12$.
3. Forgetting final $\div K_e$ in Walter (whole bracket).
4. Gordon denominator non-positive when $br \geq K_e$ — flag " $K_e > br$ violated."
5. MM's D_1 cancellation is the point — different firm values = arithmetic slip.
6. MM new shares issued at P_1 , not P_0 .
7. "Bonus increases wealth" — false; only packaging changes.
8. Higher dividend raises price only when $r < K_e$; for growth firm it *lowers* price.
9. Bird-in-hand is Gordon's, NOT MM's (MM reject it).
10. MM irrelevance is conditional on assumptions, not "always."
11. D_0 vs D_1 in Gordon: "just paid" → grow by $(1+g)$; "expected" → use as-is.
12. g is often not given — derive $g = b \times r$; $b = 1 - \text{payout}$.
13. Bonus (capitalises reserves, raises paid-up capital, face value same) vs split (re-denominates only).
14. Residual theory explains *how much* is paid, not a price — don't compute a price from it.
15. Buyback's EPS rise ≠ wealth creation — cite tax/signalling/flexibility.
16. Legal ceiling (Sec 123) — carried-forward losses may bar payout.

One-line recall

- $r > K_e \rightarrow$ retain (0% payout); $r < K_e \rightarrow$ distribute (100%); $r = K_e \rightarrow$ indifferent.
- Walter = level perpetuity, corner solution; Gordon = growing perpetuity, guardrail $K_e > b_r$, $g = b_r$.
- MM: homemade dividends + arbitrage $\rightarrow D_1$ cancels $\rightarrow$ value set by investment policy, not payout.
- Gordon reversed = dividend-growth cost of equity: $K_e = D_1/P + g$ (grow D_0 first).
- Bonus/split leave wealth unchanged; buyback's EPS rise is concentration, real gains are tax/signal/flexibility.
- Real world = between poles: sticky dividends (Lintner), residual after capex, buybacks as flexible tax-smart route.

Working Capital Management

Snapshot

Profit is annual and optional; solvency is daily and non-negotiable. A profitable firm can be **solvent but illiquid** and fail on a Tuesday. WCM = sizing the money frozen in the operating pipeline, shrinking it, and funding it.

Net Working Capital = Current Assets – Current Liabilities = (Inventory + Receivables + Cash) – (Payables + short-term dues).

Three-way distinction: **profitability** (income statement) ≠ **liquidity** (near-term) ≠ **solvency** (long-term). Technical insolvency kills before balance-sheet insolvency appears.

Core concepts

- **Liquidity–profitability trade-off:** more current assets = safety, lower ROI (ROI = Profit ÷ Investment; WC in denominator). Conservative (safe, low ROI) / Moderate (matching) / Aggressive (risky, high ROI).
- **Gross WC** = total current assets (treasurer's view); **Net WC** = CA – CL (financier's view).
- **Permanent WC** (always-full core, a rising staircase — funded long-term) vs **Fluctuating/temporary WC** (seasonal spikes, self-liquidating — funded short-term).
- **Cycle differs by industry:** trading light (no WIP), heavy engineering huge WIP, fast-food/transport can be negative (paid before paying suppliers), service = big debtors no inventory.
- **Determinants:** nature of business, scale, production cycle length, seasonal fluctuations, credit policy given/received, growth, operating efficiency/turnover, inflation, RM availability, dividend/depreciation policy.

Key provisions / rules

Operating cycle (denominators differ — the #1 slip)

Leg	Formula (days)	Denominator flow
RM holding (R)	$\text{Avg RM stock} \div (\text{Annual RM consumption} \div 365)$	RM consumption
WIP holding (W)	$\text{Avg WIP} \div (\text{Annual cost of production} \div 365)$	Cost of production
FG holding (F)	$\text{Avg FG stock} \div (\text{Annual COGS} \div 365)$	COGS
Debtors (D)	$\text{Avg debtors} \div (\text{Annual credit sales} \div 365)$	Credit sales
Creditors (C)	$\text{Avg creditors} \div (\text{Annual credit purchases} \div 365)$	Credit purchases

- **Gross Operating Cycle** = R + W + F + D
- **Cash Conversion Cycle** = R + W + F + D – C
- Cycles per year = $365 \div \text{cycle days}$; **WC** ≈ **annual cash operating cost ÷ (365 ÷ cycle days)** (cross-check).
- **Consumption = Opening stock + Purchases – Closing stock** (RM leg uses consumption; creditors leg uses purchases — back out purchases if only consumption given).
- Use average of opening/closing when both given; else state assumption.

Estimation (line-by-line)

- Current assets: RM (consumption × days ÷ 365); **WIP (RM at 100%, labour & overheads at % completion, often 50%)**; FG (COGS × days); Debtors (at cost of sales OR sales per instruction); cash float; prepaid/advances to suppliers (add).
- Less spontaneous liabilities: creditors (purchases × days), outstanding wages (wages × lag), outstanding overheads, advances received from customers (deduct).
- Net WC = CA – CL; add contingency margin only if asked.
- **Cash-cost basis:** strip non-cash depreciation from FG and debtors valuation; leave creditors/wages untouched.
- **% of sales method** = quick first-cut (WC ≈ % × forecast sales); blind to efficiency changes.

Inventory

Item	Formula
EOQ	$\sqrt{2 \times A \times O \div C}$ — A=annual usage, O=order cost, C=carry cost/unit/yr
At EOQ	total ordering cost = total carrying cost
Orders/year	$A \div \text{EOQ}$; average inventory = $\text{EOQ} \div 2$ (+ safety stock)
Re-order level	Max usage $\times$ Max lead time
Minimum level	Reorder level – (normal usage $\times$ normal lead time)
Inventory turnover	$\text{COGS} \div \text{average inventory}$

Also: ABC analysis (vital few A vs trivial many C), safety stock, JIT, quantity discounts (compare total annual cost = purchase + ordering + carrying at EOQ vs each discount qty).

Receivables (credit policy): relax credit only if **extra contribution from extra sales > extra debtor-carrying cost + extra bad debts + extra discounts**. Opportunity cost charged on **investment in debtors (at cost), not sales value**. Levers: credit period, cash discount (e.g. 2/10 net 30), credit standards (5 Cs — Character, Capacity, Capital, Collateral, Conditions), collection effort. Monitor: ageing schedule, average collection period.

Payables (spontaneous, free until discount forgone):

- **Cost of forgoing discount** $\approx [d \div (100 - d)] \times [365 \div (\text{credit period} - \text{discount period})]$
- Take discount if this annualised cost > short-term borrowing rate; else keep free credit. Denominator uses *actual delay* if firm stretches beyond net date.
- Spontaneous finance (trade credit, outstanding wages) vs negotiated (cash credit, overdraft, CP — always priced).

Cash: three motives — transaction, precautionary, speculative.

Model	Formula	Use
Baumol (steady)	$C^* = \sqrt{2 \times b \times T \div i}$	predictable drain (EOQ twin)
Miller-Orr spread	$Z = 3 \times \sqrt[3]{(3 \times b \times \sigma^2 \div 4i)}$	volatile two-way flows
Miller-Orr return point	Lower limit + (Spread $\div 3$)	
Miller-Orr upper limit	Lower limit + Spread	
Baumol: at C^* , transaction cost = interest forgone; conversions/year = $T \div C^*$; avg cash = $C^*/2$. Float toolkit: speed collection (lock-box, e-transfer), slow disbursement (pay on due date), concentration banking.		

Financing — matching (hedging) principle: fund asset with source of matching maturity. Permanent WC $\leftarrow$ long-term (equity, debentures, term loan, retained profit); fluctuating WC $\leftarrow$ short-term (cash credit, overdraft, bill discounting, CP, trade credit). Short cheaper but carries **refinancing risk + interest-rate risk**.

- Short-term sources: trade credit (spontaneous), cash credit/overdraft, bill/invoice discounting, **factoring** (recourse = firm bears bad debt, cheaper / non-recourse = factor bears it, dearer), commercial paper (high-rated only), public deposits.
- **Working-capital gap (MPBF) = Current assets – current liabilities other than bank borrowing**. Tandon Method I: bank funds 75% of gap (borrower 25%). Method II: borrower funds 25% of *total current assets* (stricter, target current ratio $\approx 1.33:1$). Method III: strictest (core CA also long-term-funded). Verify exact % / AY.

Worked mini-example

Estimation with WIP nuance (Ganga Products): 60,000 units. RM ₹80, labour ₹30, OH ₹30, SP ₹160. Annual RM 48,00,000; labour 18,00,000; OH 18,00,000; total cost 84,00,000. Norms: RM 1 mth; WIP 0.5 mth (RM 100%, conversion 50%); FG 1 mth; Debtors 2 mth at cost; creditors 1 mth; wages lag 0.5 mth; +10% contingency; cash ₹1,50,000.

- RM stock = $48,00,000 \times 1/12 = 4,00,000$
- WIP: $RM\ 48,00,000 \times 0.5/12 \times 100\% = 2,00,000$; Labour $18,00,000 \times 0.5/12 \times 50\% = 37,500$; OH = 37,500
- FG = $84,00,000 \times 1/12 = 7,00,000$; Debtors = $84,00,000 \times 2/12 = 14,00,000$; Cash = 1,50,000
- **Total CA = 29,25,000**
- Creditors = $48,00,000 \times 1/12 = 4,00,000$; Wages = $18,00,000 \times 0.5/12 = 75,000$; **CL = 4,75,000**
- Net WC = 24,50,000; +10% = 2,45,000; **WC required = ₹26,95,000** (If debtors at sales: $96,00,000 \times 2/12 = 16,00,000$, net WC +₹2,00,000.)

Cost of forgoing discount (2/15 net 45, borrow 18%): $[2/98] \times [365/30] = 0.020408 \times 12.1667 = 24.8\% > 18\% \rightarrow$ **take the discount.** (If firm stretches to day 60: $[2/98] \times [365/45] = 16.5\% < 18\% \rightarrow$ forgo discount, pay day 60.)

Exam traps & must-remember

1. **Wrong denominator per stage** (RM→consumption, WIP→cost of production, FG→COGS, debtors→sales, creditors→purchases). Using "sales" for all = #1 error.
2. WIP conversion cost scaled to completion; RM in WIP at 100% (don't halve RM).
3. Debtors at cost vs sales — read the instruction, state assumption.
4. Debtor/creditor periods use **credit** (not total) sales/purchases.
5. **Consumption vs purchases** in creditors leg — back out purchases if needed.
6. 365 vs 360 vs 12×30 days — use as specified, don't mix mid-solution.
7. Contingency margin — apply only if asked, on net WC.
8. Cash-cost basis: strip depreciation from FG & debtors; leave creditors/wages.
9. Cost of forgoing discount — take discount when its cost **exceeds** borrowing rate; denominator = (net period – discount period) or (actual delay – discount period) if stretched.
10. Opportunity cost on debtors **at cost**, not sales.
11. Baumol (steady) vs Miller-Orr (volatile) — naming wrong model loses conceptual marks.
12. Square-root damping: 4× rise in order/transaction cost only **doubles** the optimum.
13. Keep gross/net and permanent/fluctuating pairs straight.
14. Trade credit free only until discount forgone (then 20–40% implied).
15. MPBF % — verify current norms; rely on principle (bank funds only the gap).

One-line recall

- Profit is annual and optional; solvency is daily and non-negotiable.
- $CCC = R + W + F + D - C$; each leg = avg balance ÷ (annual flow ÷ days); match numerator valuation to denominator flow.
- WIP: RM 100%, conversion at % completion; debtors at cost or sales per instruction.
- $EOQ = \sqrt{2AO/C}$; Baumol $C^* = \sqrt{2bT/i}$; at optimum the two costs are equal; 4× input → 2× optimum.
- Take a cash discount when $[d/(100-d)] \times [365/(\text{credit} - \text{discount period})] > \text{borrowing rate}$.
- Matching principle: permanent WC ← long funds, fluctuating WC ← short funds; MPBF = CA – non-bank CL.

Chapter 10 — SM: Introduction to Strategic Management

Snapshot

Strategy is the child of scarcity, uncertainty, and rivalry (plus irreversibility of big commitments). FM asks "are we doing the thing right?" (efficiency); SM asks "are we doing the right thing?" (effectiveness). Strategy directs scarce resources toward a desirable competitive future, under uncertainty, while making a crowd of people pull one way — through commitments that are hard to reverse. Core chain: Vision → Mission → Objectives → Strategy → Tactics, with Values guarding every rung; three nested levels (corporate/business/functional); the prize is competitive advantage; the will is strategic intent; the machine is a four-stage loop.

Core concepts

Strategy (Chandler, 1962): determination of the basic long-term goals and objectives + adoption of courses of action + allocation of resources to carry them out. Bundles goals + action + resources.

- **Strategy is NOT:** forecasting (predicts vs decides/commits); a to-do list/budget (operational); efficiency. It IS effectiveness and **trade-offs** — choosing to be different means choosing what NOT to do.
- Two lenses (Mintzberg): **intended/deliberate** (planned) vs **emergent/realised** (pattern that actually shows up).

Strategic Management: set of managerial decisions and actions determining an organisation's long-run performance. Four activities: environmental scanning, formulation, implementation, evaluation & control.

- **Three enduring questions:** Where are we now? / Where do we want to go? / How do we get there?
- **Benefits:** direction; prepares for future/reduces surprise; rational allocation of scarce resources; builds competitive advantage; coordination & control; makes firm proactive not reactive.
- **Limitations:** turbulent/unpredictable environment; time- and cost-intensive; rigidity can blind to change; hard to forecast accurately; implementation harder than formulation; false certainty/over-commitment; demands top-management time and skill.
- **SM vs Strategic Planning:** planning = formulation only (analyse, forecast, produce the plan); management = formulation + implementation + evaluation & control, continuous and firm-wide. Planning stops at the plan; management carries it through.

The hierarchy of purpose (top-down for meaning; bottom-up for feasibility via the means–ends chain — each level is an end for the one below, a means for the one above). Horizons telescope/nest.

Rung	Question	Horizon	Nature
Vision	What to become?	Very long (aspirational)	Inspirational, future
Mission	What business now / why exist?	Present, ongoing	Identity
Objectives/Goals	What results, by when?	Medium (1–5 yrs)	Measurable targets
Strategy	How deploy resources vs rivals?	Medium–long	Method
Tactics/Plans/Policies	What daily actions?	Short	Operational

- **Values** = ethical guard-rails constraining every rung simultaneously; a real value costs a profit to uphold (else it is a slogan).
- **Vision:** desired future state — clear, inspiring, forward-looking, memorable, stretching; NOT measurable; sets a boundary that rejects off-course ventures. A pure forecast is not a vision.
- **Mission (ICAI content):** covers basic product/service & markets; customers; principal technology; philosophy/self-concept/public image. Feasible, precise, clear, motivating, distinctive. Broad enough to grow, specific enough for identity. Define by **need served, not product made** (Levitt's marketing myopia — railroads vs "transportation") — but don't over-broaden (Goldilocks band).
- **Objectives vs Goals:** goals = broad, qualitative, open-ended direction; objectives = specific, measurable, time-bound milestones. Objective characteristics: specific & measurable; realistic yet challenging; time-bound;

consistent & hierarchical; understandable & acceptable. Dark side: tunnel vision, gaming/short-termism, conflicting objectives — fix via balance & consistency.

Key frameworks / models

Mintzberg's full vocabulary: Intended (planned) · Deliberate (intended AND implemented) · Emergent (arose unplanned) · Realised = deliberate + emergent · Unrealised (intended but abandoned). Line: "Realised strategy is rarely the same as intended; the difference = unrealised intentions dropped + emergent patterns picked up."

Mintzberg's 5 Ps: Plan (intended course) · Ploy (manoeuvre to outwit a rival) · Pattern (consistency over time = emergent) · Position (locating firm in market) · Perspective (firm's ingrained way of seeing).

Levels of strategy (classify by SCOPE, not department keyword):

Level	Decided by	Question	Concern
Corporate	Top mgmt/board	Which businesses to be in?	Portfolio, scope, diversification, resource split across units (parenting)
Business (SBU)	SBU heads	How to win in THIS market?	Competitive advantage in one industry
Functional	Dept heads	How does each function support?	Resource productivity + alignment within a function

- **SBU** = distinct division with its own mission, own competitors, own market/P&L, plannable independently. Levels must nest (lower serves higher). **Single-business firm → corporate and business levels coincide.**

Competitive advantage: superior value to customers, or comparable value at lower cost, in a way rivals find hard to imitate → above-average returns.

- **Porter's generic routes:** cost leadership · differentiation · focus (either route in a niche). Must choose one — a firm being both cheapest and most differentiated is "**stuck in the middle**" and earns below-average returns.
- **Sustainable advantage** = protected by barriers to imitation. **VRIN/VRIO:** Valuable (helps exploit opportunity/neutralise threat) · Rare (else only parity) · Inimitable & non-substitutable (makes it last) · Organised to exploit / Non-substitutable (fourth condition, both formulations exist). Insight: Valuable + Rare gives advantage; Inimitable makes it last; Organised lets you capture it. Fast-moving industries: sustainable asset shifts from product to a renewal capability (string of temporary advantages).

Strategic intent (Hamel & Prahalad, 1989): obsessive, long-term ambition to achieve a leadership position, beyond current resources, driving innovation and capability-building. Expressed THROUGH vision + mission + objectives. Characteristics: captures essence of winning; stable over time; sets a stretch target; engages emotions/energy of all employees. Examples: Canon vs Xerox, Komatsu vs Caterpillar.

- **Fit vs Stretch:** Fit (traditional) = start from current resources, choose strategies they support (efficient within limits; self-limiting). Stretch/intent = start from ambition beyond means, leverage resources to close the gap (expands limits). Fit suits resource-rich mature firms; stretch suits resource-poor challengers.
- **Resource leverage (closes the gap):** Concentrate · Accumulate · Complement · Conserve · Recover. Intent without a leverage plan is fantasy.

Strategic management process (a LOOP, not a line):

1. **Environmental/Strategic Appraisal (Scanning)** — external O/T + internal S/W (SWOT, PESTLE live here). "Where are we now?"
2. **Strategy Formulation** — set vision/mission/objectives + design corporate/business/functional strategies. "Where do we go?" *Effectiveness.*
3. **Strategy Implementation** — resources, structure, systems, policies, leadership, culture, change. **Where most strategies fail.** *Efficiency.*
4. **Evaluation & Control** — set standards, measure, correct → **feeds back into Stage 1.**

Formulation vs Implementation:

	Formulation	Implementation
Nature	Positioning before action	Managing during action
Focus	Effectiveness (right things)	Efficiency (things right)
Activity	Intellectual, analytical	Operational, motivational
Skills	Analysis, judgement	Leadership, coordination
Requires	Few strategists at top	Many people across firm

Deep point: a mediocre strategy well implemented usually beats a brilliant strategy poorly implemented.

Strategy vs Tactics: strategy = overall, long-horizon, top-level, hard to reverse ("win the war"); tactics = short-horizon, specific, lower-level, adjustable ("win each battle").

Strategic vs Operational decision: strategic = long-horizon, resource-committing & largely irreversible, top-management, directional/holistic (sets the frame), environment-facing, complex/non-routine. Operational = short, reversible, low-level, repetitive, works within the frame. Test: does it commit big resources over a long horizon in a way that shapes the arena for smaller choices?

Exam traps & must-remember

- **Vision ↔ Mission swap:** Vision = future "become"; Mission = present "business/why." Anything measurable/dated = **objective** (never a vision, e.g. "₹800 cr by 2028").
- **Strategy ≠ long-term planning** (planning assumes knowable future; strategy handles uncertainty + emergent). And SM ≠ strategic planning.
- **Effectiveness vs efficiency:** strategy/formulation = right things; operations/implementation = things right. Both can appear in one correct answer.
- **Levels: classify by SCOPE, not the department name.** "Marketing" can be corporate, business, or functional.
- **Always show the feedback loop** in the process (line diagram loses marks).
- **Strategic intent ≠ vision** — intent is the stretching will to win, expressed through vision/mission/objectives; attribute to Hamel & Prahalad; stress stretch + resource leverage.
- **Name the author** — ICAI rewards attribution.
- **Give benefits AND limitations** when asked to "discuss."
- **Values are not decoration** — name them; a real value would cost a profit.
- **Don't over-apply Levitt** — too-broad mission loses identity (Goldilocks band).
- **Don't force three levels on a single-business firm** (top two coincide).
- **Spot emergent strategy** — divergence from plan due to market opportunity is emergent, not a planning failure; realised = deliberate + emergent.
- **Not every big-spend decision is strategic** — test horizon + irreversibility + frame-setting.
- **Author roster:** Chandler (strategy definition; structure follows strategy) · Mintzberg (deliberate vs emergent; 5 Ps) · Porter (competitive advantage/generic strategies/five forces/stuck-in-the-middle) · Hamel & Prahalad (strategic intent, resource leverage, core competence) · Levitt (marketing myopia).
- **Answer format (4–6 marks):** name concept + author + why it exists + one-line example.

One-line recall

- Strategy = scarcity + uncertainty + rivalry + irreversibility → one deliberate, adjustable direction; it is trade-offs (what NOT to do).
- Hierarchy: Vision (become) → Mission (present business) → Objectives (measurable/dated) → Strategy → Tactics; Values guard every rung.
- Three nested levels — corporate (which businesses), business/SBU (how to win here), functional (how each dept supports); classify by scope.

- Competitive advantage = valuable + hard-to-imitate → above-average returns; sustainable via VRIN; routes = cost leadership/differentiation/focus, avoid stuck-in-the-middle.
- Strategic intent (Hamel & Prahalad) = stretch ambition beyond resources, closed by resource leverage; Fit (match to means) vs Stretch (grow the means); realised = deliberate + emergent (Mintzberg).
- Process = appraise → formulate → implement → evaluate, feeding back; formulation chooses right things, implementation (where most fail) does them right.

Chapter 11 — SM: Analysis of the External Environment

Snapshot

Opportunities and threats arise ENTIRELY OUTSIDE the firm; strengths and weaknesses are inside. Strategy = matching the firm to an uncontrollable environment. Environmental scanning (map-making step) converts raw external facts into just two usable outputs: opportunities to exploit and threats to defend against. Anything that becomes neither is noise. Scanned in layers, far to near: macro (PESTLE) → industry (Porter's Five Forces) → strategic groups → individual competitor. Use a telescope for the far layer, a microscope for the near.

Core concepts

Environmental scanning — process of monitoring, evaluating and disseminating information from the external (and internal) environment to key people. Purpose: EARLY WARNING of threats and EARLY SIGHTING of opportunities, so the firm responds before change fully arrives.

Two failure modes (symmetrical):

- Under-scanning (blind spot) → missed threat/opportunity, strategic surprise.
- Over-scanning (analysis paralysis) → cost, delay, inertia.
- Correct: scan SELECTIVELY — monitor strategically relevant environment intensely, the rest lightly.

What a scan looks for (event → trend → issue → expectation ladder — a maturity sequence; read early):

1. Events — important specific occurrences (new tax, product launch).
2. Trends — general direction events move in.
3. Issues — concern/debate arising from trends.
4. Expectations — demands of interested groups/stakeholders.

Environment split:

- Micro (task/operating) — actors close to firm: suppliers, customers, competitors, market intermediaries, financiers, public. Firm-specific, partly influenceable.
- Macro (general/remote) — broad forces on ALL firms: PESTLE. Uncontrollable, adapt only.

Three tests to place a factor (macro vs micro):

1. "Everyone" test — affects all firms = macro; specific to your sector/firm = micro.
2. "Controllability" test — can only adapt = macro; can influence = micro.
3. "Named actor" test — impersonal force (technology, demographics) = macro; named actor you transact with (supplier, buyer) = micro.

Internal vs external is a separate (orthogonal) cut: internal factors → Strengths/Weaknesses; external (macro+micro) → Opportunities/Threats. Trap: an internal factor ("skilled workforce") is a strength, NEVER an opportunity.

Four scanning modes (broad-cheap → deep-costly): undirected viewing → conditioned viewing → informal search → formal search. Escalate selectively.

Key frameworks / models

PESTLE (macro checklist — coverage, not calculation; evolved from Aguilar's ETPS → PEST → PESTLE; STEEPLE adds Ethical)

Letter	Factor	Covers / signals
P	Political	Govt stability, ideology, centre-state relations, taxation, foreign-trade rules, "Make in India", govt attitude — the rules of the game / intent

Letter	Factor	Covers / signals
E	Economic	GDP, inflation, interest & exchange rates, disposable income, credit, business cycle, fiscal & monetary policy — size/health of demand, cost of capital
S	Social/Socio-cultural	Demographics (age, urbanisation), lifestyle, education, attitudes, tastes, values, family structure — what people want and how many; slow, deadly if missed
T	Technological	Innovation rate, R&D, automation, new methods, obsolescence, digital disruption — obsolescence threat + new-product opportunity; fastest force
L	Legal	Company/competition/consumer/labour/environmental law, contracts, IPR, product safety — codified enforceable rules; compliance risk
E	Environmental/Ecological	Climate, pollution norms, carbon regulation, resources, sustainability, waste rules — green constraints and opportunities

- Political vs Legal: Political = govt's STANCE/INTENT/discretion (what it wants/might do); Legal = CODIFIED enforceable rules already on statute book (what law already requires). "Govt is pro-manufacturing" = Political; "must comply with Factories Act" = Legal.
- How to use: list factors per head → assess direction & impact → classify O or T FOR THIS FIRM → judge timing/importance → feed into SWOT.
- Two-axis screen (importance × probability): High impact + high probability = act now; high impact + low probability = contingency plan; low impact + high probability = monitor; low + low = ignore.
- Same fact, opposite verdict (never state O/T without naming the firm): rising interest rates — threat to developer, opportunity to bank/NBFC; ageing population — threat to toymaker, opportunity to pharma; weak rupee — threat to importer, opportunity to IT exporter; drought — threat to rural FMCG, opportunity to irrigation-pump maker.

Porter's Five Forces (industry attractiveness — Michael Porter, 1980, *Competitive Strategy*)

Industry profit is a value pool leaking in five structural directions: upstream (suppliers), downstream (buyers), sideways (rivals), capped (substitutes), diluted (new entrants). Four forces press in on the central rivalry.

- **Threat of new entrants** — bring capacity, want share, cut prices. Held back by BARRIERS TO ENTRY: economies of scale, capital requirements, product differentiation/brand loyalty, switching costs, access to distribution, government policy/licensing, cost advantages independent of scale (patents, tech, location, experience curve), expected incumbent retaliation. High barriers → low threat → high profit. It is the THREAT (contestability), not actual entry, that disciplines prices.
- **Bargaining power of suppliers** — powerful when: few/concentrated, input critical/no substitute, high switching costs, can threaten forward integration, industry is unimportant customer. Value leaks upstream. LABOUR (employees/unions) counts as a supplier.
- **Bargaining power of buyers** — powerful when: few/large-volume, product standardised, low switching costs, buyers earn low profits (price-sensitive), can threaten backward integration, product unimportant to their quality or large share of their cost. Needs BOTH price sensitivity AND bargaining leverage.
- **Threat of substitutes** — products from OTHER industries meeting the same need (tea vs coffee, video-conf vs air travel). Place a CEILING on price. High when better price-performance + low switching cost. Most-forgotten force; includes "do nothing" and "make in-house".
- **Competitive rivalry (central force)** — intense when: many/balanced competitors, slow growth, high fixed/storage costs, no differentiation/low switching costs, capacity added in large increments, diverse competitors, high strategic stakes, HIGH EXIT BARRIERS. Worst structural cell = HIGH ENTRY + HIGH EXIT barriers (chronic overcapacity, price wars — steel, airlines).

Golden rule: strong forces → LOW profit; weak forces → HIGH profit. Attractive industry = WEAK forces. Strategy: (a) position where forces weakest, (b) exploit change first, (c) reshape forces (raise switching costs, build scale). Read the TREND, not just the snapshot.

Limitations: static snapshot; understates cooperation/complementors; assumes clear industry boundaries; under-weights firm's unique resources (RBV critique). Sixth force = COMPLEMENTORS (Andrew Grove) — opposite of a substitute, raises your product's value; cooperative strategy. ICAI = classical FIVE forces; complementor is an extension only.

Competitor analysis (Porter — four components; motivation × ability matrix)

- **Future goals** — what drives it (growth/cash/prestige); content or hungry.
- **Current strategy** — what it does now; baseline to out-maneuvre.
- **Assumptions** — beliefs about itself/industry; wrong assumptions = exploitable BLIND SPOTS (highest-value, hardest to observe).
- **Capabilities** — genuine strengths/weaknesses; limits on response.

Goals + Assumptions = motivation/perception; Strategy + Capabilities = action/ability. Answers: Is rival satisfied? What moves will it make? Where vulnerable? What provokes fierce retaliation? Intelligence sources must be LEGAL/ETHICAL (financials, filings, teardowns, job postings, patents, statements — no espionage).

Strategic groups

Clusters of firms following SIMILAR strategies (price/quality, product-line breadth, geographic reach, distribution, vertical integration). Strategic group MAP plots firms on two dimensions.

- Closest rivals = firms WITHIN your group; rivalry fiercest within a group.
- MOBILITY BARRIERS block movement between groups (like entry barriers inside an industry).
- Empty cells = POSSIBLE white space — but check WHY empty (may be structurally unprofitable).
- Five forces press on different groups with different intensity.
- Choose two axes that are strategically important AND uncorrelated (else firms line up on a diagonal, revealing nothing).

Exam traps & must-remember

1. Describing instead of implying — always convert PESTLE factor / force into firm-specific O/T or profit effect.
2. Forgetting substitutes (most-missed force); include "not buying" and "in-house".
3. Rivalry vs substitutes: rivalry = SAME product (Pepsi vs Coke); substitutes = DIFFERENT product, same need (juice vs cola).
4. New entrants (potential/future, kept out by barriers) vs existing rivals (current competition).
5. Profitability logic: strong forces = LOW profit; weak = HIGH. Attractive = weak forces (students invert this).
6. PESTLE Political (stance/intent) vs Legal (enforceable laws); PESTLE is MACRO — suppliers/customers/competitors are micro (Five Forces, not PESTLE).
7. Environment has no fixed meaning — same factor is O for one firm, T for another ("recession threat — to whom?").
8. Authors: Five Forces, generic strategies, competitor 4-components, strategic groups = Michael Porter (1980). PESTLE ← Aguilar's ETPS. Do not credit Ansoff/Mintzberg.
9. High barriers PROTECT incumbents and DETER entrants (reduce the threat) — don't reverse direction.
10. Firm can RESHAPE forces — not a fatalistic verdict.
11. Never put an internal factor (workforce, brand, balance sheet) into the external scan — those are S/W.
12. Read dynamics/trend, not just a snapshot (deregulation, consolidation, new tech move the forces).
13. Weight forces unequally — identify the DOMINANT one (prioritisation = depth signal).
14. Empty strategic-group cell may be structurally unprofitable, not overlooked white space.
15. Substitute (caps your price) vs complementor (raises your value) — opposite directions.

Connections: External analysis = the O and T half of SWOT (internal = S and W). Scanning = stage two of the strategic-management process (after vision/mission/objectives, before strategy formulation). Five Forces → leads into Porter's generic strategies (cost leadership, differentiation, focus) as the RESPONSES. Strong forces justify a higher discount rate / conservative forecasts in FM capital budgeting.

One-line recall

- O and T come from OUTSIDE; S and W from inside — never swap them.
- Layers, outer to inner: PESTLE → Five Forces → strategic groups → competitor analysis, each ending in an O/T or profit implication.

- PESTLE = completeness checklist (P-E-S-T-L-E); Political = intent, Legal = codified law.
- Five Forces (Porter 1980): new entrants, suppliers, buyers, substitutes, rivalry — STRONG forces = LOW profit; worst cell = high entry + high exit barriers.
- Competitor analysis = goals + strategy + assumptions + capabilities (motivation × ability); attack the wrong assumption.
- Analysis is worthless until converted into a strategic implication — an opportunity to seize or a threat to defend.

Chapter 12 — SM: Analysis of the Internal Environment

Snapshot

External analysis tells you what is *attractive*; internal analysis tells you what is *achievable*. Two firms facing the same environment pick different strategies because of what lies inside them. This chapter is the **Resource-Based View (RBV)** half of strategy: advantage comes from *what you own* (resources rivals cannot copy), not just *where you sit* (positioning school). The two schools are **complements, not rivals** — the RBV exists because positioning could not explain why two firms in identical positions earn differently. Payoff = the *inside-out* view of sustainable competitive advantage.

Three sub-questions: (1) What do we have? (2) Which of these is genuinely special? (3) Where in our activities is value created or destroyed?

Core concepts

Resource = any asset (tangible or intangible) a firm owns/controls and can *deploy* to conceive and execute strategy. Definition = deployability toward strategy, NOT accounting recognition (reputation is a resource but off the balance sheet).

Three families:

- **Tangible** — physical, on balance sheet: plant, machinery, land, cash, raw-material stock, distribution hardware.
- **Intangible** — non-physical, often off balance sheet: brand/reputation, patents & trademarks, proprietary technology, culture, customer relationships, know-how.
- **Human** — skills/knowledge in people: managerial talent, engineering expertise, worker productivity, tacit skills.

Key point: **intangibles are usually the stronger source of advantage** — invisible, hard to value, cannot be bought, must be built over years; that inimitability is what makes them precious. **Financial resources (cash)** are a legitimate tangible resource (deployable to buy others) but rarely a source of advantage — capital is mobile and non-rare ⇒ usually **Valuable but not Rare = parity**.

Capability (competence) = capacity to deploy resources in a *coordinated* way to a desired end. Resides in routines, processes, organisational glue — not a single asset. Built by learning and repetition; cross-functional. Resources = nouns; capabilities = verbs; advantage lives in the verbs. An idle resource earns nothing.

- **Threshold capabilities** — minimum needed just to be in the game (bank clears cheques). Ceiling = parity; absence = fatal weakness.
- **Distinctive / core capabilities** — let the firm *outperform* rivals. Advantage lives only here. Mistake = over-investing in threshold capabilities.
- **Dynamic capability** — higher-order ability to *reconfigure, renew, rebuild* ordinary capabilities as the environment shifts (sense, seize, transform). Answer to the core-rigidity trap. If a stem says "turbulent / rapidly changing environment," this is the concept being tested.

Core rigidity — yesterday's core competence hardens into today's trap when the world changes (Kodak film chemistry blocked its digital pivot). The *more* inimitable the competence, the *harder* to abandon. So internal analysis is a continuous audit, never a one-time verdict.

Key frameworks / models

Core competence — Prahalad & Hamel (HBR, 1990), 3 tests

1. **Access to a wide variety of markets** — transferable, underpins many products (Honda engines → cars, bikes, generators, mowers).
2. **Significant contribution to perceived customer benefits** — customers value it.
3. **Difficult for competitors to imitate** — complex, tacit, built over time.

Tree metaphor: **core competencies = roots; core products = trunk; business units = branches; end products = leaves/fruit.** Think of the firm as a *portfolio of competencies*, not just businesses. "Core" = strategic label (central + rare + inimitable + transferable), NOT a synonym for "essential/important" (payroll is important, not core).

Ladder of narrowing scope:

Term	Meaning	Test
Competence / capability	Any coordinated ability to deploy resources	Just needs to exist
Core competence	Central, transferable, valued, hard to imitate	Prahalad & Hamel's 3 tests
Distinctive competence	Core competence performed <i>better than all rivals</i>	Core + superiority
Competitive advantage	Superior value/profit the market rewards	Above-average returns

Anchor: *a core competence becomes distinctive when it beats rivals; a distinctive competence becomes competitive advantage when the market rewards it.*

VRIO / VRIN — Jay Barney (analytical heart of RBV)

Original = **VRIN** (Valuable, Rare, Inimitable, Non-substitutable); Barney later reframed the 4th filter as **Organisation** ⇒ **VRIO**. Cumulative gates — must clear each in turn:

Filter	Question	If NO ⇒
V — Valuable	Exploits opportunity / neutralises threat, improves efficiency or effectiveness?	Competitive disadvantage
R — Rare	Held by only a few (ideally one) rivals?	Competitive parity
I — Inimitable (costly to imitate)	Would rivals face significant cost disadvantage to obtain/develop it?	Temporary advantage
O — Organised to exploit (VRIO) / N — Non-substitutable (VRIN)	Firm has structure/systems/processes to capture value? / No equivalent substitute exists?	Unrealised / substitutable

All pass ⇒ **Sustained competitive advantage.**

Why inimitability is the linchpin (isolating mechanisms):

- **Unique historical conditions** — path in time that cannot be rerun (first-mover, founder's vision).
- **Causal ambiguity** — even the firm cannot articulate why it is good; rivals can't copy what nobody can specify (culture).
- **Social complexity** — rests on interpersonal relationships, trust, reputation.

O = "so what — can you actually use it?" Structure, controls, culture, rewards must convert potential into realised advantage. **N** guards against a rival reaching the *same customer benefit by a different route* (inimitable canal beaten by a railway). Modern practice: quote **VRIO** primary, name **non-substitutability** if the question raises substitutes.

Action per verdict:

Verdict	Exits at	Do
Disadvantage	Fails V	Fix, outsource, or exit
Parity	Passes V, fails R	Maintain at efficient cost; don't over-invest
Temporary advantage	Passes V,R; fails I	Exploit fast; build a durable asset before rivals copy
Unrealised advantage	Passes V,R,I; fails O	Fix structure/systems, not the resource
Sustained advantage	Passes all	Protect and deepen

V is not permanent — re-test against the *current* environment. Kodak's film mastery stayed rare and inimitable but the V-gate flipped to No ⇒ straight to disadvantage. Formal RBV statement of core rigidity.

Value Chain — Michael Porter (Competitive Advantage, 1985)

VRIO says *whether* you have advantage; the value chain says *where* it comes from. Disaggregates the firm into **nine strategically relevant activities**. **Margin = total value created – total cost of activities**.

Primary activities (create & deliver the product):

Activity	Covers
Inbound logistics	Receiving, storing, distributing inputs; material handling, warehousing, inventory
Operations	Transform inputs into product; machining, assembly, packaging
Outbound logistics	Store & distribute finished goods; order processing, delivery
Marketing & sales	Inform buyers, induce purchase; advertising, pricing, channels, sales force
Service	Maintain/enhance value after sale; installation, repair, training, spares

Support (secondary) activities:

Activity	Covers
Firm infrastructure	General management, planning, finance, accounting, legal, quality
Human resource management	Recruiting, training, developing, compensating personnel
Technology development	R&D, process automation, product design, know-how
Procurement	The <i>function</i> of purchasing inputs (distinct from inbound logistics = physical handling)

How to use: (1) identify each activity + cost; (2) diagnose sources of advantage; (3) examine **linkages** (interdependencies — better inspection in Operations cuts Service cost later; one fix heals two wounds); (4) benchmark vs rivals' chains. The chain sits inside a larger **value system** (suppliers ↔ firm ↔ channels/buyers); advantage often lives in managing interfaces (JIT).

Two lenses on one chain: *cost drivers* (scale, capacity utilisation, linkages, location, timing) for **cost leadership**; *uniqueness drivers* for **differentiation**. Cannot maximise both in the same activity ⇒ "stuck in the middle" warning. **Margin ≠ accounting profit** — it is buyer's willingness-to-pay minus total activity cost. For **service firms**, keep the nine categories but *re-interpret* them (Operations = core service delivery; inbound/outbound light or informational).

SWOT and TOWS

- **Strengths / Weaknesses** = *internal* (output of resource/capability/competence/value-chain analysis). Test = ownership + origin (does the firm control it, inside its boundary), NOT good-vs-bad.
- **Opportunities / Threats** = *external* (arise in the environment, apply to all players).

Worked test: "government EV subsidy" = **Opportunity** (external, open to all); "our EV battery patents" = **Strength** (internal, ours). A fact can be a strength AND a weakness at once (deep specialisation = world-class competence + dangerous dependence/rigidity).

SWOT is a *list*, not a *strategy*. **TOWS Matrix** — **Heinz Wehrich** (name reverses SWOT) *pairs* internal with external to generate options:

	Strengths (S)	Weaknesses (W)
Opportunities (O)	SO — Maxi-Maxi: strengths seize opportunities (aggressive growth)	WO — Mini-Maxi: overcome weaknesses to grab opportunities (turnaround)
Threats (T)	ST — Maxi-Mini: strengths counter threats (defend/diversify)	WT — Mini-Mini: minimise weaknesses, avoid threats (defensive/retrench)

SWOT limitations: static snapshot; unweighted/unprioritised; same item argued into different boxes; long lists that paralyse. TOWS fixes "list not strategy" but still doesn't rank which family to pursue first.

Framework roll-call (author + tool)

Framework	Author	Answers
Resource-Based View	Wernerfelt; developed by Barney	<i>Why</i> look inside
Core competence	Prahalad & Hamel (1990)	Which capabilities <i>define</i> the firm
VRIO / VRIN	Jay Barney	<i>Whether</i> a resource yields sustainable advantage
Value Chain	Michael Porter (1985)	<i>Where</i> value & cost arise
SWOT	Traditional (Learned/Andrews)	<i>Collate</i> internal + external
TOWS Matrix	Heinz Wehrich	<i>Combine</i> into strategies

Examinable pipeline

Resources (tangible/intangible/human) → Capabilities & competencies → Core competence (Prahalad-Hamel 3 tests) → VRIO/VRIN screen → Value chain map (locate advantage & leaks) → List S & W → Combine with O & T in TOWS → feasible strategy. **Self-check: every strategy must trace back to a specific VRIO-passing resource matched to a real external factor.**

Exam traps & must-remember

1. **Resource vs capability** — resource = thing you *have*; capability = something you can *do*. "Skilled engineers" (resource) vs "design a car in 18 months" (capability).
2. **Not every strength is an advantage** — most are only *parity*. Always run through VRIO first.
3. **Imitable resource ⇒ only temporary advantage** (patents expire, tech reverse-engineered). State the *type* of advantage.
4. **Never drop the O** in VRIO — un-exploited advantage yields nothing.
5. **VRIN vs VRIO** — N = Non-substitutability; O = Organisation. Barney's later, more-cited version = VRIO.
6. **Procurement (support = buying function) ≠ Inbound logistics (primary = physical handling).**
7. **Firm infrastructure, HRM, Technology development, Procurement = support**, even Technology development in a tech firm (develops know-how; Operations uses it).
8. **Don't stop at SWOT** — marks are in the TOWS SO/ST/WO/WT combination.
9. **Don't swap TOWS cells** — SO = Maxi-Maxi (aggressive); WT = Mini-Mini (defensive).
10. **Attribute the right author** — Core competence = Prahalad & Hamel; VRIO/VRIN = Barney; Value Chain = Porter; TOWS = Wehrich.
11. **Internal vs external misfiling** — test = origin & control, not favourable/unfavourable. A subsidy = external Opportunity; your patent on it = internal Strength.
12. **"Core" ≠ "important"** — must be central + rare + inimitable + transferable (passes the 3 tests).
13. **"Valuable" is not permanent** — re-test V against current environment; disruptive tech stem ⇒ former strength fails V and demands dynamic capabilities.
14. **Threshold vs distinctive** — a threshold capability can never be an edge no matter how well done.
15. **Don't force Porter's manufacturing chain on a service firm** — re-interpret the nine activities.
16. **Margin ≠ P&L gross margin** — it is willingness-to-pay minus total activity cost.

One-line recall

- Same industry, different profits ⇒ the gap is internal ⇒ RBV; positioning (outside) and resource-based (inside) are complements.
- Resources (tangible/intangible/human) → capabilities → core competencies (roots of the tree); intangibles are the stronger source of advantage.
- VRIO cumulative gates: fail V = disadvantage, fail R = parity, fail I = temporary, fail O/N = unrealised; all pass = sustained advantage. Inimitability from unique history, causal ambiguity, social complexity.
- Value chain = 5 primary (Inbound → Operations → Outbound → Marketing & sales → Service) + 4 support (Infrastructure, HRM, Technology dev, Procurement); Value – Cost = Margin; watch linkages.
- SWOT collates internal S/W + external O/T; TOWS (Wehrich) combines into SO/ST/WO/WT — the matching, not the listing, is the strategy.

- Advantage is perishable: keep re-testing V and hold the dynamic capability to rebuild the resource base. Inside-out view.

SM Chapter 13 — Strategic Choices

Snapshot

Finite resources force a firm to choose a direction — every "yes" is many implicit "no"s (strategy = what NOT to do). Two nested questions: (1) Corporate level = scope (which businesses; grow/hold/shrink); (2) Business level = positioning (how to beat rivals in one industry). Below sits (3) Functional level (how each department implements the business choice). The three levels must be vertically consistent. Frameworks are graded by a single risk-vs-familiarity gradient: the more that stays known, the lower the risk.

Core concepts

Three levels of strategy: Corporate (which businesses) → Business (how to compete in each) → Functional (marketing/finance/operations/HR/R&D support the chosen way of competing). A "marketing head" question = functional. Must be vertically consistent (differentiation + starved R&D = contradiction).

Four grand corporate directions:

- **Stability** — maintain current position; least risky; NOT "doing nothing" (a conscious status-quo choice; firm still improves and defends). Danger = complacency. Sub-types: no-change (settled cruising, monitor); profit/harvesting (sacrifice long-term for short-term profit, often pre-exit); pause/proceed-with-caution (deliberate temporary rest AFTER fast growth, to consolidate before growing again — a comma, not a full stop).
- **Expansion (Growth)** — substantial growth in products/markets/scale. Reasons: survival (static share erodes in growing industry), profitability & drive, prestige/psychological (empire-building — dangerous), economies of scale. Trap: uncontrolled growth beyond capacity ("over-trading") causes later distress.
- **Retrenchment** — pull back to a defensible core; chosen under distress; NOT defeat (surgery to save the patient). Escalating forms:
 - **Turnaround** — stay in business, radically overhaul to reverse decline (intention = recovery). Elements: change leadership, redefine focus, divest/curtail unwanted assets, improve profitability of retained ops, careful complementary acquisitions. Warning signs: negative cash flow, falling margins/share, adverse debt-equity, deteriorating working capital, high employee turnover, mounting inventory.
 - **Divestment** — sell/spin off a unit as a going concern (failed turnaround often ends here).
 - **Liquidation** — close down, sell assets piecemeal; last resort; LEAST attractive (admits failure, damages reputation, destroys going-concern value). Divestment preferred whenever a buyer exists.
- **Combination** — different postures across different units, simultaneous or sequential (e.g. grow software, hold steady in steel, retrench telecom). Only available to multi-business/multi-region/multi-product firms. If a case gives several SBUs different treatment → label = combination.

Expansion modes (separate axis from Ansoff direction): organic; merger; acquisition/takeover; joint venture (jointly-owned new entity, share risk / local partner abroad); strategic alliance (looser, non-equity); internationalisation (really market development elevated by added political/currency/cultural risk). Direction (which products/markets) vs mode (alone vs partner) are two different axes.

Diversification (4th Ansoff cell):

- **Related (concentric)** — new business shares a link (technology, production skills, distribution, customers, brand). Logic = synergy ($2+2=5$). Lower risk, more defensible.
- **Unrelated (conglomerate)** — no operational link; enter attractive industry. Logic = financial (spread risk, deploy surplus cash, general management skill). Higher risk. Often value-destroying.
- Four synergy types: **Marketing** (brand, distribution, sales force, advertising); **Operating** (shared plants, technology, procurement, spread fixed costs); **Investment** (shared plant, machinery, R&D, inventories, tooling); **Management** (know-how transferred to similar problems).
- Sharp critique: pure risk-spreading is NOT a valid corporate justification — investors diversify more cheaply themselves (conglomerate discount = head-office cost, cross-subsidy, opacity). Diversification must add what the

investor cannot replicate.

Integration (subset of expansion — grow along industry's own value system):

- **Vertical** — into own supply chain. Backward/upstream = towards supplier (secure inputs, control quality, capture margin). Forward/downstream = towards customer (control route to market, capture retailer margin, customer info).
- **Horizontal** — acquire/merge same-stage rivals (market share, scale, less competition, wider line).
- Costs/risks: raises fixed costs & reduces flexibility (locked into in-house supplier); demands new competences (can be unrelated diversification in disguise); dulls incentives (captive supplier complacent); balancing problem (upstream/downstream scales mismatch). Counter-trend = outsourcing / de-integration. Horizontal integration triggers competition (anti-trust) law scrutiny — India: Competition Commission clearance above thresholds ("appreciable adverse effect on competition").

Key frameworks / models

Ansoff Product-Market Matrix (Igor Ansoff, 1957) — where to grow, rising risk:

	Existing market	New market
Existing product	Market Penetration (lowest risk)	Market Development
New product	Product Development	Diversification (highest risk)

- Penetration: sell more of same to same (usage frequency, win rivals' customers, convert non-users, price, advertising). Has a natural ceiling → forces firms up the gradient.
- Market Development: existing product to new geographies/segments/channels/uses (NOT just "going abroad").
- Product Development: new products/variants to existing loyal customers.
- Diversification: new on both axes, highest risk.
- Market vs product development both "moderate"; relative risk depends on where the firm's competence and environment's uncertainty lie.

Porter's Generic Strategies (Michael Porter, 1980) — how a single business beats rivals; 2 choices: source (low cost vs differentiation) × scope (broad vs narrow):

	Broad market	Narrow market
Low cost	Cost Leadership	Cost Focus
Uniqueness	Differentiation	Differentiation Focus

- **Cost Leadership** — lowest-COST producer (not necessarily lowest price; cost is the weapon). Sources: economies of scale, tight cost control, efficient ops, low-cost inputs, standardisation, experience curve. Defends vs 5 forces: survive price wars (rivals); buyers push only to next-efficient rival's floor; volume gives supplier power; scale = entry barrier; low price beats substitutes. Requirement: product must stay acceptable/comparable to buyers.
- **Differentiation** — perceived unique across industry (quality, design, brand, features, service, technology) → price premium → loyalty, lower price sensitivity. Economic condition: profitable ONLY if premium buyers will pay EXCEEDS extra cost of distinctiveness. Failure modes: over-differentiation (more than buyers pay for); uniqueness buyers can't perceive/verify. Vulnerable to imitation and downturns.
- **Focus (Niche)** — narrow segment served exceptionally well, via cost focus OR differentiation focus (not both). Works because broad rivals serve the "average" customer, leaving segments under- or over-served. Evaporates if niche needs converge with mainstream or niche too small.
- **Stuck in the Middle** — try both cost + differentiation broadly → achieve neither → below-average profit (beaten on price by cost leader, on value by differentiator; underlying activities structurally conflict). Exception (Porter conceded): a few firms briefly achieve both (e.g. process innovation cheaper AND better) — usually temporary. A focuser can also get stuck-in-the-middle within its niche.

BCG Growth-Share Matrix (Boston Consulting Group, 1970) — which multi-business units to fund/milk/cut. Axes: market growth rate (vertical = cash NEEDED, high/low line $\approx 10\%$) $\times$ relative market share (horizontal = cash GENERATED, line at $1.0\times$).

- **RMS = SBU's own share $\div$ largest rival's share.** >1.0 = leader (high side); <1.0 = follower. Reversing the ratio flips every quadrant.

Quadrant	Growth / Share	Strategy
Star	Hi / Hi	Invest to hold leadership (tomorrow's cash cow)
Cash Cow	Lo / Hi	Milk; surplus funds Stars & Question Marks
Question Mark (Problem Child)	Hi / Lo	Invest selectively to build into Star, or divest
Dog	Lo / Lo	Divest or liquidate

- Cash-flow story: cows fund stars & selected question marks; cut dogs. Dynamic: success sequence (QM $\rightarrow$ Star $\rightarrow$ Cash Cow); disaster sequence (Star $\rightarrow$ QM $\rightarrow$ Dog, or over-milked cow leaves only dogs). Balance across the life cycle, not just today's cash.
- Limitations: only 2 crude variables; share alone drives profitability; "market" ambiguous; "Dog" over-condemns.

GE/McKinsey Nine-Cell Matrix (GE & McKinsey, early 1970s) — same decision, richer axes. Industry (market) Attractiveness (growth, size, profitability, competitive intensity, regulation) $\times$ Business (competitive) Strength (share, brand, cost position, technology, quality). Each axis = weighted composite (weights sum to 1.00; rate each factor 1–5; weight $\times$ rating, sum). 3×3 grid scored High/Med/Low.

- Green zone (attractive + strong) = Invest/Grow; yellow diagonal = Selectivity/Hold/Earn; red (weak + unattractive) = Harvest/Divest.
- BOTH axes must be high to justify aggressive investment (an attractive industry alone does not). Weakness: subjective weighting.

Feature	BCG	GE/McKinsey
Cells	4 (2×2)	9 (3×3)
Vertical axis	Market growth (single)	Industry attractiveness (composite)
Horizontal axis	Relative market share (single)	Business strength (composite)
Driver	Cash flow / experience curve	Attractiveness $\times$ ability to compete
Strength	Simple, vivid cash logic	Richer, finer resolution
Weakness	2 crude variables; Dog over-condemns	Subjective weights; judgement-heavy

Exam traps & must-remember

1. Stability $\neq$ doing nothing (active status-quo choice). Distinguish pause (rest before more growth) from no-change (settled cruising).
2. Don't confuse levels: "compete against a rival" $\rightarrow$ Porter; "which businesses to invest in" $\rightarrow$ BCG/GE; marketing manager's strategy $\rightarrow$ functional.
3. Cost leadership = lowest COST, not necessarily lowest price; product must stay acceptable.
4. Integration (own value system) $\neq$ diversification (new business). Backward/forward = vertical; same-stage = horizontal. Backward into a very different upstream can be unrelated diversification in disguise.
5. Ansoff risk order: Penetration (lowest) $\rightarrow$ Market/Product Development (moderate) $\rightarrow$ Diversification (highest). Market development includes new segments/channels/uses, not just abroad.
6. "Stuck in the middle" = trying to be both cost leader AND differentiator broadly $\rightarrow$ neither (activities conflict); not merely "a weak firm." Focusers can get stuck too.
7. BCG axes: vertical = growth (cash needed); horizontal = relative market share (cash generated). Dog = low-growth AND low-share.
8. RMS = own share $\div$ LARGEST rival's share, line at $1.0\times$. Not raw share, not rival $\div$ yours. 18% own share can be a low-share follower (rival 30% $\rightarrow 0.60\times$).

9. BCG (4 cells, single-factor axes) vs GE (9 cells, weighted multi-factor composites). "Weights/ratings/nine cells" → GE.
10. Related diversification's rationale = synergy (name the type); unrelated = financial. Don't attribute operational synergy to unrelated diversification.
11. Retrenchment ≠ failure. Order: turnaround → divestment → liquidation (least attractive, last resort).
12. Attractive industry alone does NOT justify investment on GE — own business strength must also be high (else Selectivity/Hold, not green).
13. BCG is dynamic, not static (success vs disaster sequence). A portfolio of only cash cows with no stars/question marks is strategically bankrupt.
14. Diversification is one Ansoff cell / one expansion mode, not a synonym for growth.

One-line recall

- Corporate directions (4): Stability, Expansion, Retrenchment (turnaround → divestment → liquidation), Combination.
- Ansoff (1957): Penetration → Market Dev → Product Dev → Diversification (rising risk); related diversification = synergy (marketing/operating/investment/management).
- Porter (1980): Cost Leadership / Differentiation / Focus; refuse to choose = stuck in the middle. Differentiation pays only if premium > added cost AND uniqueness survives imitation.
- BCG (1970): Star (invest) / Cash Cow (milk) / Question Mark (selective/exit) / Dog (divest); RMS = own ÷ largest rival, line 1.0x; cows fund stars & question marks.
- GE/McKinsey (early 1970s): Industry Attractiveness × Business Strength (weighted composites), 9 cells; BOTH must be high to invest aggressively.
- Spine: finite resources force a choice → corporate scope + business positioning → refusing to choose loses.

SM Chapter 14 — Strategy Implementation & Evaluation

Snapshot

- A brilliant strategy that is not executed is worth less than a mediocre one that is. Most strategies fail in execution, not choice — the **strategy-execution gap**. Failure occurs at the *weakest link*, not the average.
- Four managerial questions → four tool families: resistance → change management (Lewin); wrong org chart → structure follows strategy (Chandler); culture/skills pull old way → McKinsey 7S; flying blind → strategic control + Balanced Scorecard.
- Formulation = judgement under uncertainty (once); implementation = coordination under interdependence (continuous). Implementation, control and evaluation run **simultaneously** as overlapping loops — strategy is a cycle, not a line. Strategic management is NOT a one-time exercise.

Core concepts

Formulation vs Implementation (memorise the contrast):

Dimension	Formulation	Implementation
Nature	Positioning forces <i>before</i> action	Managing forces <i>during</i> action
Focus	Effectiveness (right things)	Efficiency (things right)
Process	Intellectual / analytical	Operational / administrative
Skills	Analytical, conceptual	Motivational, leadership, integrating
People	Few, at the top	Many, across all levels

- They are *interrelated but not identical* — analytically separable, operationally intertwined. Implementation reality can force reformulation.
- **Forward linkage:** strategy design shapes execution (bold strategy → more change). **Backward linkage:** existing structure/systems/resources constrain which strategies are feasible.
- **Implementation activities:** institutionalising the strategy; resource allocation; structuring; behavioural implementation (leadership, culture, change); functional/operational implementation (plans, policies, procedures per function).
- **Six barriers to implementation:** (i) vague communication; (ii) resources still fund old priorities; (iii) rewards tied to old behaviour; (iv) structure misfits strategy; (v) leadership talks but doesn't model change; (vi) culture contradicts strategy.

Key frameworks / models

1. Lewin's Three-Stage Change Model

1. **Unfreeze** — create motivation; challenge status quo ("burning platform"); reduce forces holding old behaviour.
 2. **Move / Change** — implement new behaviours, structures, systems, skills; training, quick wins, support.
 3. **Refreeze** — lock in via revised rewards, SOPs, culture, structure. Change not refrozen decays.
- **Force-field logic:** behaviour held by *driving forces* (push change) vs *restraining forces* (hold status quo). Leverage = *reduce restraints*, not crank up driving force (that only raises tension and rebounds).
 - **Sources of resistance:** fear of unknown; loss of self-interest (pay/status/power/security); distrust of management; different perceptions; social/peer pressure; low tolerance for change. Diagnose source before choosing tactic.
 - **Kotter & Schlesinger — six methods (slow/durable → fast/resented):** (1) Education & communication (misinformation); (2) Participation & involvement (initiators lack info, others can resist — builds ownership); (3) Facilitation & support (fear/adjustment); (4) Negotiation & agreement (someone clearly loses & has power); (5) Manipulation & co-optation (covert, symbolic role — backfire risk); (6) Coercion (force/threat — last resort). Move *down* the list only as speed pressure and power imbalance rise.
 - **Kotter 8-step = operationalises Lewin:** steps 1-4 (urgency, coalition, vision, communicate) = Unfreeze; steps 5-7 (empower/remove obstacles, short-term wins, consolidate) = Move; step 8 (anchor in culture) = Refreeze. Step 6

short-term wins keep momentum.

- Resistance reduced most by **participation** — people support what they help create.

2. Structure Follows Strategy (Chandler)

- Studied DuPont, GM, Sears, Standard Oil: change strategy → old structure misfits → inefficiency/administrative problems → performance declines → redesign structure → performance restored.
- **Two-way truth:** dominant proposition = "structure follows strategy" (state cleanly first); nuance = structure can subsequently *constrain* strategy (feedback, not one-way). Never lead with the nuance.
- **Structural types:**

Structure	Best fit	Key trade-off
Simple/Entrepreneurial	Small, single owner, one product	Fast/flexible; overloads owner; fails on growth
Functional	Single/dominant product, stable	Efficient/specialised; silos, poor coordination
Divisional (product/geo/market)	Diversified products/regions/customers	Accountability/focus; resource duplication
SBU	Large, highly diversified, many divisions	Groups related divisions; adds a layer
Matrix	Multiple projects needing shared specialists	Dual expertise; dual-reporting conflict
Network/Virtual/Hourglass	Outsource non-core; flat, IT-enabled	Lean/adaptive; partner dependence, less control

- **Trade-off lens:** *specialisation* (functional — experts together) vs *coordination* (divisional — everything a customer needs under one boss). Matrix chases both → two-boss problem. Derive: single product + stable → functional; many products → divisional.
- **Trigger to leave functional:** diversification-driven overload at the apex (CEO becomes bottleneck).
- **SBU characteristics:** distinct business; own set of competitors; manager accountable for strategic planning & profit. Restores span-of-control; cost = extra layer, lost cross-SBU economies.
- **Hourglass:** IT + BPR thin the middle-management layer; broad top + broad bottom, narrow middle. Benefits: lower cost, faster communication, empowered lower levels. Cost: fewer promotions, overloaded surviving middle.
- **Network/Virtual:** keep only core competence, outsource everything else via IT web. Max lean/flexible; surrenders control, partner-dependent. (Hourglass *thins* middle in-house; network *externalises* activities.)

3. Leadership & Culture

- **Strategic leadership:** anticipate, envision, stay flexible, think strategically, initiate change for a viable future.
- **Transformational** (charisma, inspirational motivation, intellectual stimulation, individualised consideration) → changes hearts → powers *Unfreeze/Move*.
- **Transactional** (contingent reward, management-by-exception) → steady execution → powers *Refreeze*.
- Leader's two roles: **change agent** (visible sponsor — vision, resources, removes obstacles) and **culture-builder** (shapes values via what is rewarded/modelled). CEO = "chief implementation officer"; missing leader-sponsor is the top predictor of failure.
- **Culture** = shared values/beliefs/assumptions/norms — "culture eats strategy for breakfast." Active variable, not backdrop. Shaped via rewarded behaviours, stories/heroes, rituals/symbols, recruitment/socialisation, reward system.
- **Strategy-culture compatibility matrix** (importance of change × compatibility with culture): (1) **Manage around** — important but mildly incompatible (reroute via structure/procedures); (2) **Reinforce** — important + compatible (easiest; ride culture); (3) **Change culture** — vital + strongly opposed (hardest, slowest, last resort); (4) **Tolerate/reformulate** — minor + incompatible.

4. McKinsey 7S (Peters, Waterman, + Pascale, Athos)

Alignment of seven interdependent elements = successful implementation; change one, realign the rest.

- **Hard (tangible, changeable by decree, fast):** Strategy (build competitive advantage); Structure (who reports to whom); Systems (daily procedures, processes, info flows — budgeting, IT, appraisal).
- **Soft (intangible, live in habits, slow, decisive):** Shared Values (originally Superordinate Goals — core beliefs/culture at the *centre*, the glue); Style (leadership/management behaviour); Staff (people — numbers, recruitment, development, motivation); Skills (organisation's collective distinctive competencies).
- **Core trap:** managers change the easy *hard* S's, declare victory, and stall because the decisive *soft* S's never moved. Excellent companies win on the soft S's.
- **Staff ≠ Skills:** Staff = people as resources; Skills = what the firm *as a whole* is distinctively good at (survives turnover). **Skills ≠ Systems:** Systems = formal procedures; Skills = competencies.
- **Diagnostic:** list current vs desired state of each S → find gaps/conflicts; start with Shared Values, check hard then soft elements against strategy, map interconnections, realign iteratively. It is an alignment loop, not a one-time checklist.

5. Strategic Evaluation & Control

Determine effectiveness of a strategy and take corrective action. Checks: (i) strategy implemented as planned; (ii) results as intended; (iii) premises still hold.

- **Difficulties of strategic control:** measurement problems (long horizons, hard attribution); environmental turbulence (moving yardstick); timing problem (results visible too late). → relies on *leading* indicators and *premise* checks.
- **Control process (6 steps):** determine what to control → set standards/benchmarks → measure actual → compare & analyse variance → judge whether deviation is *significant* (within tolerance?) → corrective action (adjust performance, revise standards, or reformulate strategy). Act only on significant deviations — over-correcting on noise destabilises.

Strategic vs Operational control:

Basis	Strategic	Operational
Focus	Direction; "right things?"	Efficiency; "things right?"
Horizon	Long term	Short term
Concerned with	Environment, assumptions, strategy	Resource use, day-to-day
Nature	External, adaptive	Internal, corrective
Data	Mostly external, qualitative	Mostly internal, quantitative
Corrective action	May reformulate strategy	Adjusts execution

Four types of strategic control (Schendel & Hofer):

1. **Premise Control** — checks assumptions/predictions (economy, industry, competitors, technology). *Focused* — each premise pre-identified and assigned.
 2. **Implementation Control** — should strategy continue given events as roll-out proceeds? Tools: monitoring strategic thrusts + milestone reviews.
 3. **Strategic Surveillance** — broad, *unfocused* monitoring of a wide range of events; catches the threat nobody listed as a premise.
 4. **Special Alert Control** — rapid reconsideration triggered by sudden, unexpected event (crisis, competitor acquisition, disaster, regulatory shock). Uses contingency plans / crisis teams.
- **Sort by 2 axes:** trigger specific/broad × monitoring continuous/event-driven. Premise = specific + continuous; Surveillance = broad + continuous; Implementation = specific + continuous (internal roll-out); Special Alert = broad + event-driven. Premise vs Surveillance discriminator = *focus* (targeted vs undirected).

6. Balanced Scorecard (Kaplan & Norton, early 1990s)

Why financial measures alone fail: lagging indicators (report past outcomes); ignore drivers of future performance (customer loyalty, process quality, capability, innovation); encourage short-termism & can be manipulated; ignore

intangible assets (brand, knowledge, relationships).

Perspective	Core question	Example measures
Financial	How do shareholders see us?	Revenue growth, ROCE, margin, cash flow
Customer	How do customers see us?	Satisfaction, retention, market share, on-time delivery
Internal Business Process	What must we excel at?	Cycle time, defect/quality rates, cost, innovation cycle
Learning & Growth (Innovation & Learning)	Can we keep improving/creating value?	Training, skills, IT capability, staff retention, new ideas

- **Causal chain (bottom-up):** Learning & Growth → Internal Process → Customer → Financial. Financials are the *last* link.
- **Four "balances":** financial/non-financial; lagging/leading; internal/external; short-term/long-term.
- Each perspective pairs a lagging *outcome* with a leading *driver* (e.g. market share vs on-time delivery).
- **Strategy map** = one-page cause-effect diagram; visual articulation of the causal chain; makes BSC a communication device.
- BSC = measurement system + strategic management & communication system.
- **Limitations:** mechanical if causal logic ignored; too many measures dilute focus; cause-effect links assumed not proven; non-financial data costly & gameable; needs continual updating.
- **Numeric BSC:** score = weight × achievement; cap achievement at 100%. *Inverse* measures (defect rate, cost, cycle time, attrition — lower is better) use **target ÷ actual**, not actual ÷ target.

Exam traps & must-remember

1. **"Structure follows strategy"** — never reverse it. State proposition first, nuance (structure later constrains strategy) second.
2. **7S** = seven elements, **Shared Values at centre**; don't list only six. Hard = Strategy/Structure/Systems; Soft = Shared Values/Style/Staff/Skills.
3. **Author-framework pairs:** Lewin = change (+ force-field); Kotter = 8-step; Kotter & Schlesinger = overcoming-resistance methods; Chandler = structure follows strategy; Peters & Waterman = 7S; Kaplan & Norton = BSC (+ strategy map); Schendel & Hofer = types of strategic control.
4. **Staff ≠ Skills ≠ Systems** in 7S — keep the three apart.
5. **BSC perspectives:** Financial, Customer, Internal Business Process, Learning & Growth (don't drop L&G or rename Internal Process). Causal direction L&G → Financial.
6. **Why financials insufficient** — give reasons (lagging, ignore drivers, short-termism, miss intangibles), not adjectives.
7. **BSC "lower is better" reversal** — inverse measures use target ÷ actual.
8. **Name all four BSC balances**, not one.
9. **Strategic vs Operational control** — practise the distinction table.
10. **Four control types** — Surveillance is *unfocused/broad*; Premise is *targeted at pre-identified assumptions*.
11. Implementation is a *set* of activities (institutionalising, resource allocation, structuring, behavioural, functional) — not one step.
12. **Lewin order:** Unfreeze → Move → Refreeze; most-forgotten = **Refreeze**.
13. **"Just change the culture"** = last resort only (vital + strongly incompatible); prefer reinforce or manage-around.
14. **Over-correcting on noise** — act only on *significant* deviations outside tolerance.

One-line recall

- Lewin: Unfreeze → Move → Refreeze; weaken *restraining* forces, don't push harder; participation is the strongest anti-resistance tactic.
- Chandler: change strategy → old structure misfits → redesign; functional (specialisation) vs divisional (coordination); diversification triggers the shift.
- 7S: align all seven around Shared Values; managers change easy hard S's and neglect the decisive soft S's.

- Control: 6-step loop acting only on significant variance; four types = Premise, Implementation, Surveillance, Special Alert (Schendel & Hofer).
- BSC: L&G → Internal Process → Customer → Financial; financials are lagging; four balances; inverse measures use $\text{target} \div \text{actual}$.
- Whole chapter = one loop: make people willing to change, give a fitting structure, align every internal element, and continuously check leading + lagging signals to correct in time.

Chapter 15 — SM: Functional & Digital Strategy

Snapshot

A corporate/business strategy is only a promise; it becomes real only when translated into concrete plans by each function (marketing, finance, operations, HR, R&D) and those plans are mutually consistent. Strategy is a cascade (down = intent/goals; up = feasibility/constraints), not a one-way memo. In the modern economy, this same discipline must extend to digital strategy — going digital is a corporate/business choice that must cascade into every function, not an IT purchase.

Four failure modes of the cascade: Translation failure (grand words never become measurable targets), Alignment failure (functions contradict each other), Resource failure (money/people/machines never actually reallocated), Feedback failure (top never hears the plan is infeasible).

Core concepts

Hierarchy of strategy (three levels — each serves and constrains the level below):

- Corporate — "What businesses should we be in?" (scope, portfolio; directs)
- Business (SBU) — "How do we win in each business?" (cost or differentiation; positions)
- Functional — "What must each department do to deliver the win?" (executes) Rule: if it changes which industry's profit pool you fish in = corporate; how you beat rivals in the same pool = business; how a department supports the fight = functional. Single-business firms collapse corporate and business into one. Below functional sits the operational/tactical layer (day-to-day schedules) — strategy allocates resources across a horizon; operations executes within an allocation already made.

Why the hierarchy exists: (1) different questions need different decision-makers (decentralisation); (2) scope/reversibility differs by level; (3) resources are committed inside functions; (4) consistency must be engineered — each function's default is to optimise its own local metric, and the sum of local optima $\neq$ global optimum; (5) technology changes constraints (digital can create new strategic options). Not writing one giant plan because of bounded rationality and information cost.

SBU (Strategic Business Unit) — three defining characteristics: (i) a single or related set of businesses plannable separately; (ii) its own set of competitors; (iii) a manager responsible for strategic planning and profit. Test: "Could it be sold off and survive as a standalone business?"

Grand strategy (master/corporate strategy) — Expansion, Stability, Retrenchment, Combination (Glueck & Jauch). Directional, not operational; each function translates it into a functional strategy. Functional strategy is a derivative of the grand strategy.

Key frameworks / models

Grand strategy → functional posture (change the grand strategy and every arrow flips):

Function	Expansion	Stability	Retrenchment
Marketing	New markets/products, aggressive promotion	Defend share, refine mix	Prune weak products, harvest
Finance	Raise capital, accept leverage, reinvest (low dividend)	Steady, moderate payout	Conserve cash, cut capex, sell assets
Operations	Add capacity, new plants/tech	Optimise existing capacity	Close/consolidate plants, cut inventory
HR	Hire/train fast, scale	Stable headcount, develop staff	Downsize, redeploy, VRS
R&D	Product development, widen range	Incremental improvement	Cut/defer projects, focus on essentials

The five functional strategies:

- Marketing (demand-side): STP (Segmentation → Targeting → Positioning) precedes the 4 Ps. 4 Ps = Product, Price, Place, Promotion (McCarthy); for services add People, Process, Physical evidence = 7 Ps (Booms & Bitner). Pricing: Skimming (high price; use when demand inelastic, novel/differentiated, imitation hard, recover R&D fast) vs Penetration (low price, volume; use when demand elastic, scale economies large, network effects, deter entry). Push (through channel, trade incentives) vs Pull (end-customer demand, advertising/branding). Must be consistent with business strategy.
- Finance (resource-supply): capital structure (debt-equity mix), financing decisions, investment/capital budgeting, dividend policy, working capital management. Enabler + scorekeeper (ROI, ROCE, EVA). Value check: $EVA = NOPAT - (Capital \times WACC)$. Growth is not automatically good — only growth earning above the cost of capital creates value.
- Operations (supply-side): capacity, process choice (job → batch → mass → continuous), facilities/layout, supply chain (make vs buy), quality (TQM, Six Sigma), inventory (JIT vs buffer). Efficiency ↔ flexibility trade-off (moving right = more efficient, less flexible). Order-winner vs order-qualifier (Hill): qualifier = minimum to be considered; order-winner = what clinches the sale. Where cost and quality are born.
- HR (people plan): manpower planning, recruitment/selection, training/development, performance appraisal & rewards, retention/culture/change management. Strategy-culture bridge; usually the binding constraint on speed (skills/values can't be installed overnight); reward system is the sharpest lever ("what gets measured and paid gets done").
- R&D (future-supply): innovation posture — first mover/pioneer vs follower/imitator; product vs process R&D; make vs buy technology; R&D intensity. First-mover advantages: technological leadership, pre-emption of scarce assets, buyer switching costs, set industry standards. Disadvantages: high R&D/market-education cost, uncertainty, free-rider imitators. Rule: pioneer where advantage is defensible (patents, network effects, standards); follow where imitation is cheap and market proven.

Three fits (alignment test):

1. Vertical fit — advances the business/grand strategy above.
2. Horizontal fit — consistent with other functions (marketing's delivery dates match operations' capacity).
3. Internal fit — sub-decisions consistent (premium price + cheap packaging = mismatch).

McKinsey 7S (Peters, Waterman & Pascale) — total-alignment diagnostic:

- Hard Ss: Strategy, Structure, Systems (tangible, easier to change).
- Soft Ss: Style, Staff, Skills, Shared Values (cultural, hard to change, decide execution).
- Shared Values at the centre (anchors the other six; change last). Managers over-invest in hard Ss, under-invest in soft — why restructurings fail. Cannot change strategy without changing supporting functional Ss.

Digital depth ladder (data → process → business model):

- Digitisation — analog information → digital data (paper records → database).
- Digitalisation — digital tech improves an existing process (online order form replaces phone).
- Digital transformation — digital reinvents the business model/value proposition (selling scooters → battery-swap-as-a-service).

Drivers of digital transformation (why adapt): (1) changing customer expectations; (2) competitive pressure/disruption from digital-natives (Kodak problem); (3) technology availability & falling cost (cloud/AI/analytics rentable); (4) data as a strategic asset; (5) globalisation/connectivity; (6) operational efficiency/agility. Deep reason: digital changes the basis of competition — old advantages become liabilities. Barriers (why hard): (i) legacy systems; (ii) cultural resistance/fear of job loss; (iii) skill gaps in data/AI talent; (iv) cybersecurity/data-privacy risk; (v) high upfront investment, uncertain payback; (vi) change-management fatigue.

E-business models:

- By parties: B2C, B2B, C2C, C2B, B2G/B2A.

- By revenue logic: e-tailer/online store (owns inventory, product margin); marketplace/platform (connects buyers-sellers, commission, network effects, neutral venue, seller's brand shows); subscription/SaaS (recurring fee); advertising/freemium; aggregator (one common brand + set quality standards, e.g. Ola/Oyo/Zomato); on-demand/gig; brokerage/transaction fee.
- Aggregator vs marketplace: both asset-light intermediaries. Marketplace = neutral venue, seller's own brand shows. Aggregator = fragmented providers under one common brand and quality standard; providers are partners not employees, aggregator doesn't own assets, sets quality/pricing, single unified brand.
- Network effects → winner-takes-most moat; new rivals face the cold-start (chicken-and-egg) problem; spend early to reach critical mass.

Emerging technologies as strategic levers (cross-functional, not IT-only):

Technology	What it does	Strategic use
Cloud computing	Rent computing on demand	Capex → opex; instant scale; global reach
Big Data & Analytics	Insight from large data	Demand prediction, personalisation, risk mgmt
AI/ML	Machines that learn & decide	Automation, forecasting, chatbots, fraud detection; learns/improves with data
IoT	Connected sensors on assets	Real-time monitoring, predictive maintenance, pay-per-use
Blockchain	Distributed tamper-proof ledger	Trust without intermediaries, supply-chain traceability
Robotics/RPA	Repetitive rule-based work	Lower cost, higher speed; RPA does NOT learn
Mobile & Social	Ubiquitous connected customers	Direct relationships, new channels, community

Why strategic: change the cost curve, change customer expectations, create compounding advantages (data/AI flywheel), destroy old advantages. RPA vs AI: RPA follows fixed rules for structured tasks (no learning); AI/ML handles unstructured/judgemental tasks and improves with data.

Exam traps & must-remember

- Confusing the three levels — "advertise on Instagram?" is functional, not corporate.
- Digital transformation ≠ buying software/an app (that's digitalisation); transformation changes the business model.
- Digitisation vs digitalisation vs digital transformation — data → process → business model.
- Forgetting horizontal fit — state cross-function consistency explicitly.
- 4 Ps are Product, Price, Place, Promotion (McCarthy) — NOT "Packaging". Services' extra three = People, Process, Physical evidence (Booms & Bitner).
- 7S soft Ss = Style, Staff, Skills, Shared Values; hard = Strategy, Structure, Systems; Shared Values at centre.
- Emerging tech is cross-functional, not R&D-only (AI serves marketing, operations, finance, HR).
- Marketplace (no inventory, commission, seller's brand) ≠ e-tailer (owns inventory, margin) ≠ aggregator (one common brand, set quality). C2B ≠ B2C — direction matters.
- Cascade is a loop, not a one-way waterfall — functions feed feasibility back up.
- "Why adapt?" answered weakly — name the strategic drivers AND barriers; deep reason = digital changes the basis of competition.
- Don't praise digital/growth without a value check — test EVA > 0 (return beats WACC).
- Two computed options (skimming vs penetration, e-tailer vs marketplace) — marks are in the strategic reasoning, not the bigger number.
- Grand strategy flips every functional arrow — don't give expansion-style plans for a retrenchment scenario.
- Rule-based automation (RPA) is not AI — RPA doesn't learn.

Connections: Functional strategy = Porter's generic strategies made operational; maps onto Porter's Value Chain (primary + support activities); Ansoff's market/product development lands on marketing/R&D; 7S links to implementation &

structure; finance-as-scorekeeper links to strategic control (Balanced Scorecard: financial, customer, internal-process, learning); BCG portfolio decisions become each SBU's financial constraint; financial strategy is the clearest SM ↔ FM bridge.

One-line recall

- Corporate directs, business positions, functional executes — cascade down (intent), feedback up (feasibility).
- Five functional strategies: Marketing (STP then 4/7 Ps), Finance (structure/budgeting/dividend/WC; EVA), Operations (efficiency ↔ flexibility, order-winner), HR (binding constraint, reward lever), R&D (pioneer vs follower).
- Three fits = vertical + horizontal + internal; 7S hard = Strategy/Structure/Systems, soft = Style/Staff/Skills/Shared Values (centre).
- Digital ladder: digitisation (data) → digitalisation (process) → digital transformation (business model).
- E-models: B2C/B2B/C2C/C2B/B2G; e-tailer vs marketplace vs aggregator; network effects → winner-takes-most.
- Change the grand strategy and every functional arrow flips; digital transformation must itself cascade and clear the cost-of-capital hurdle.